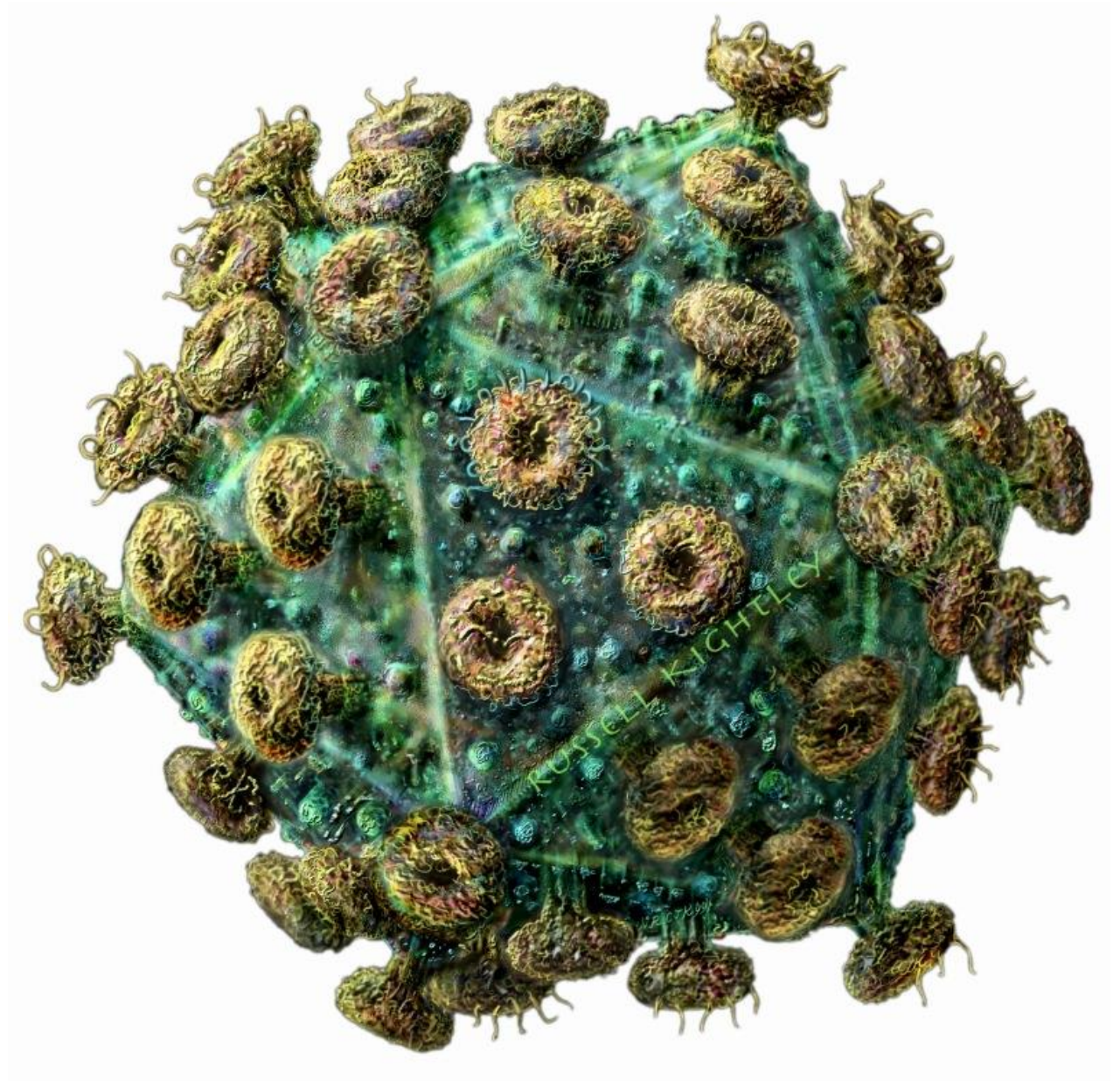
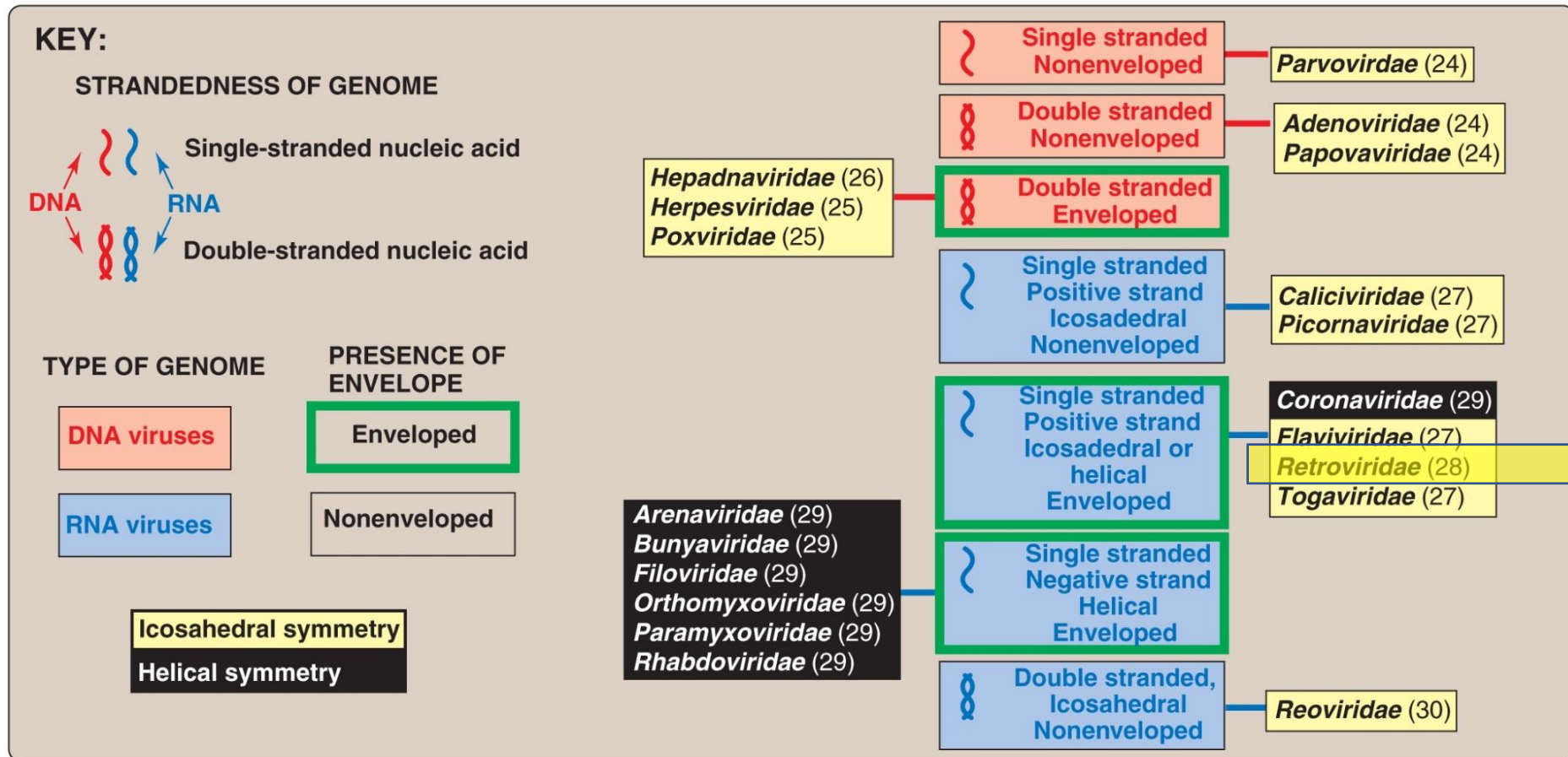


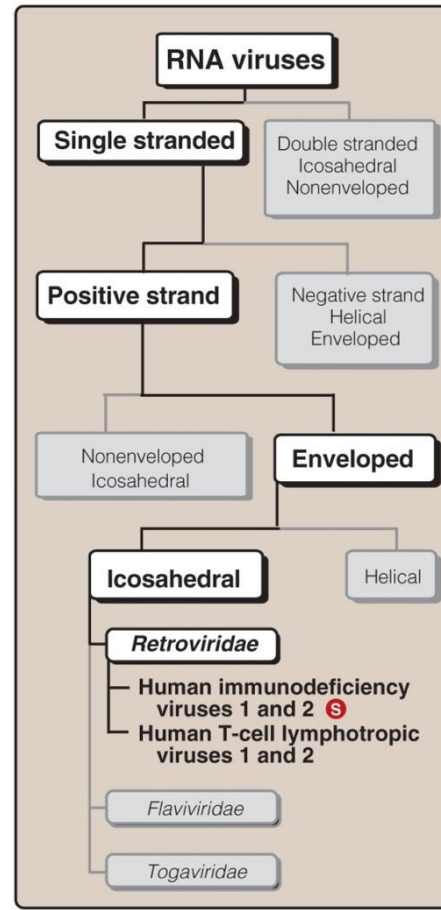
HIV

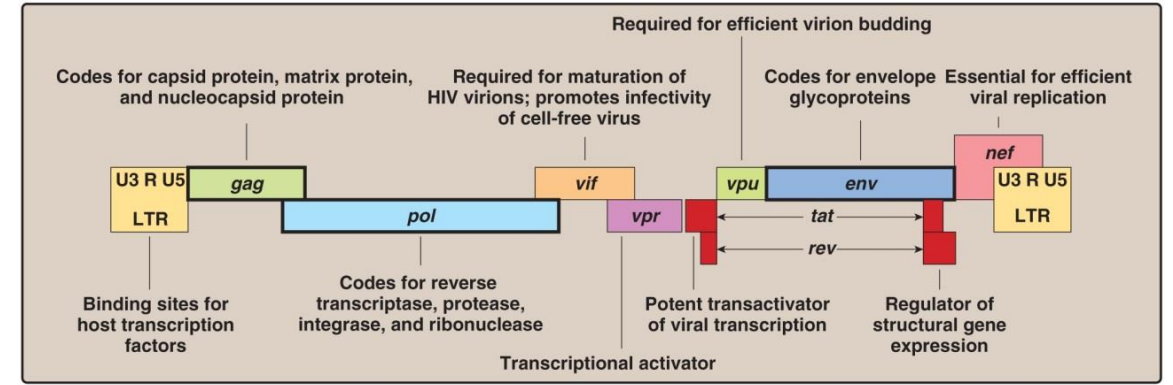
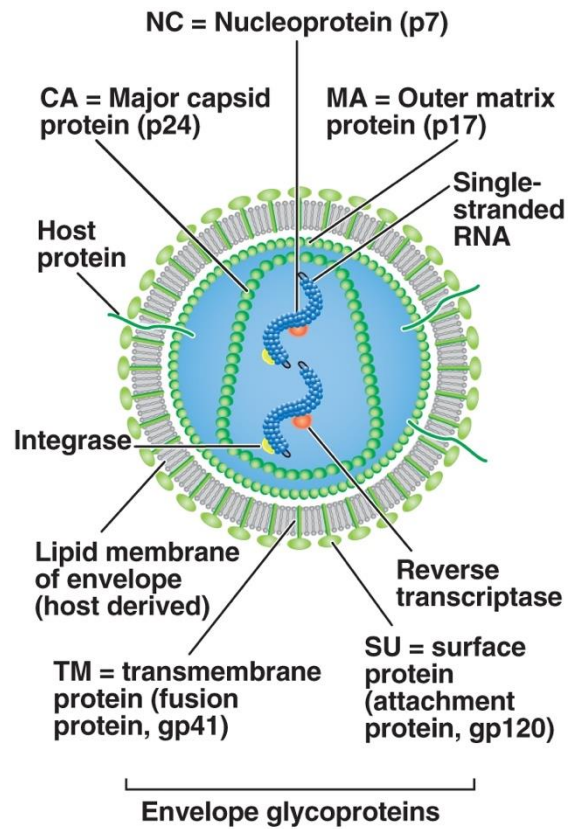


Viruses



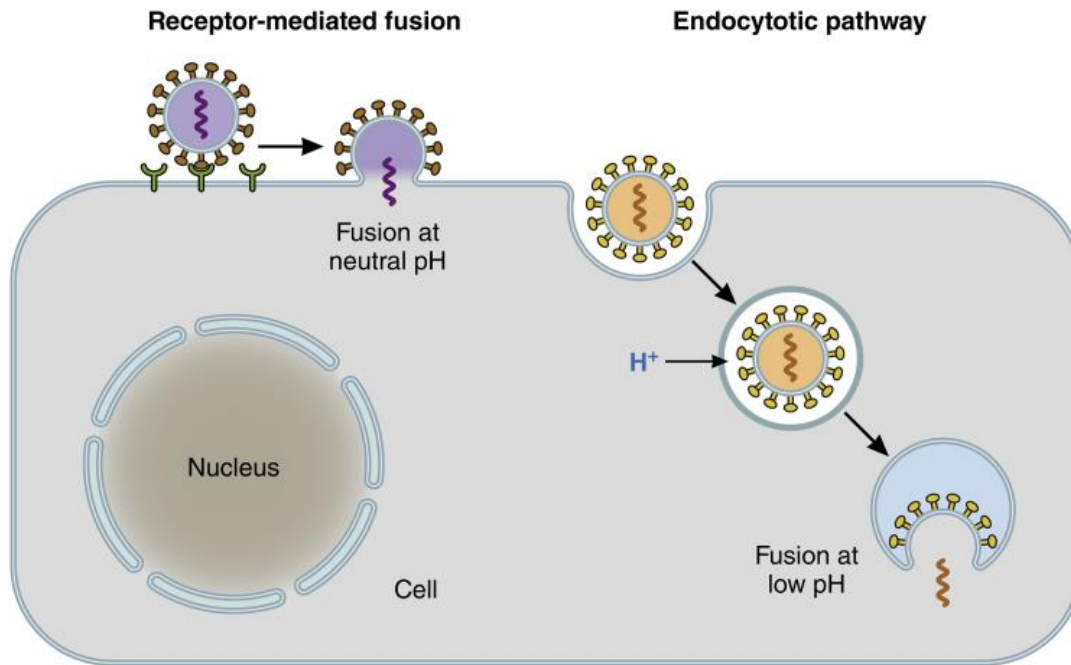
HIV



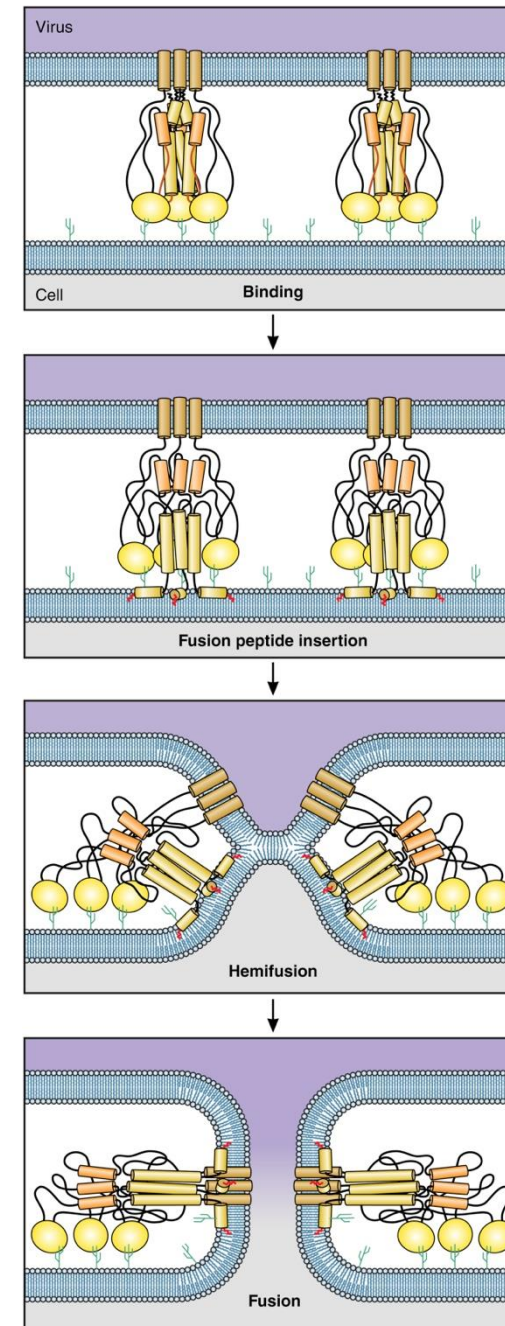


HIV virus

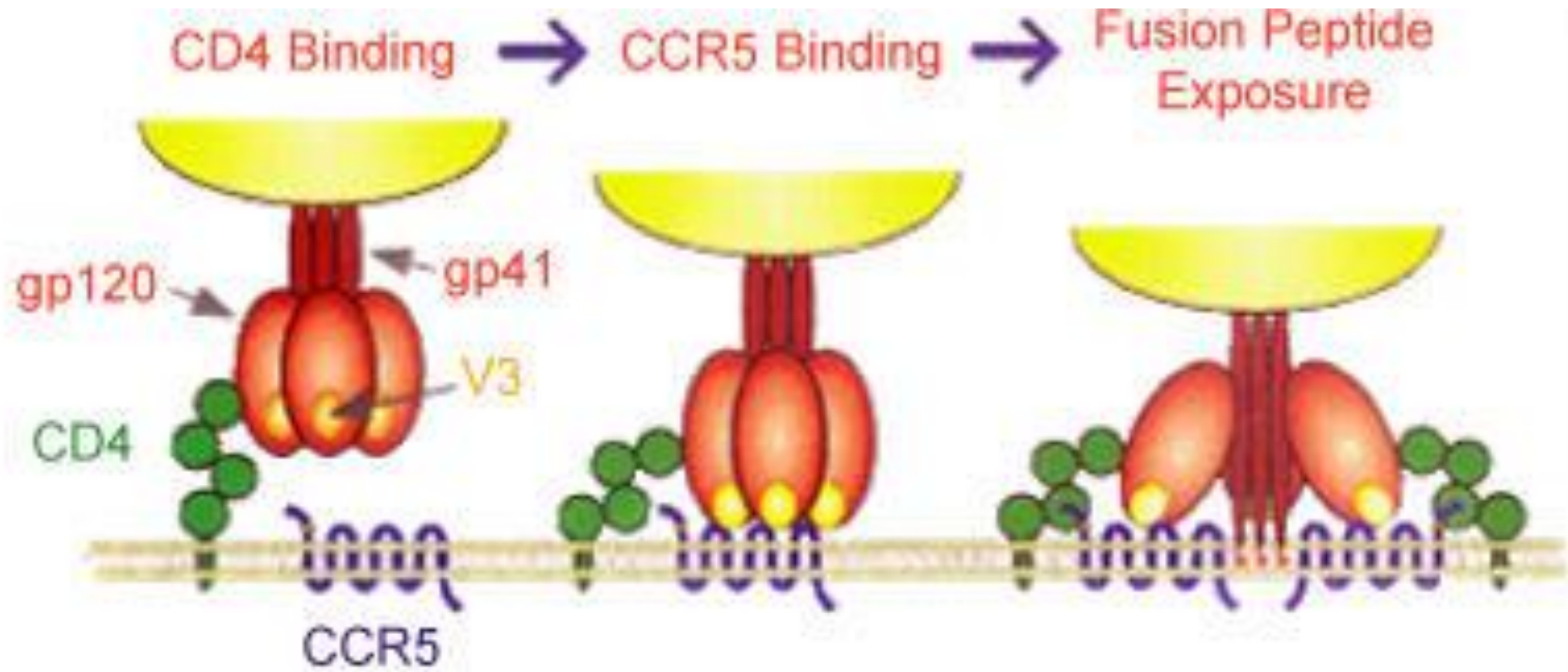
Virus entry of a cell



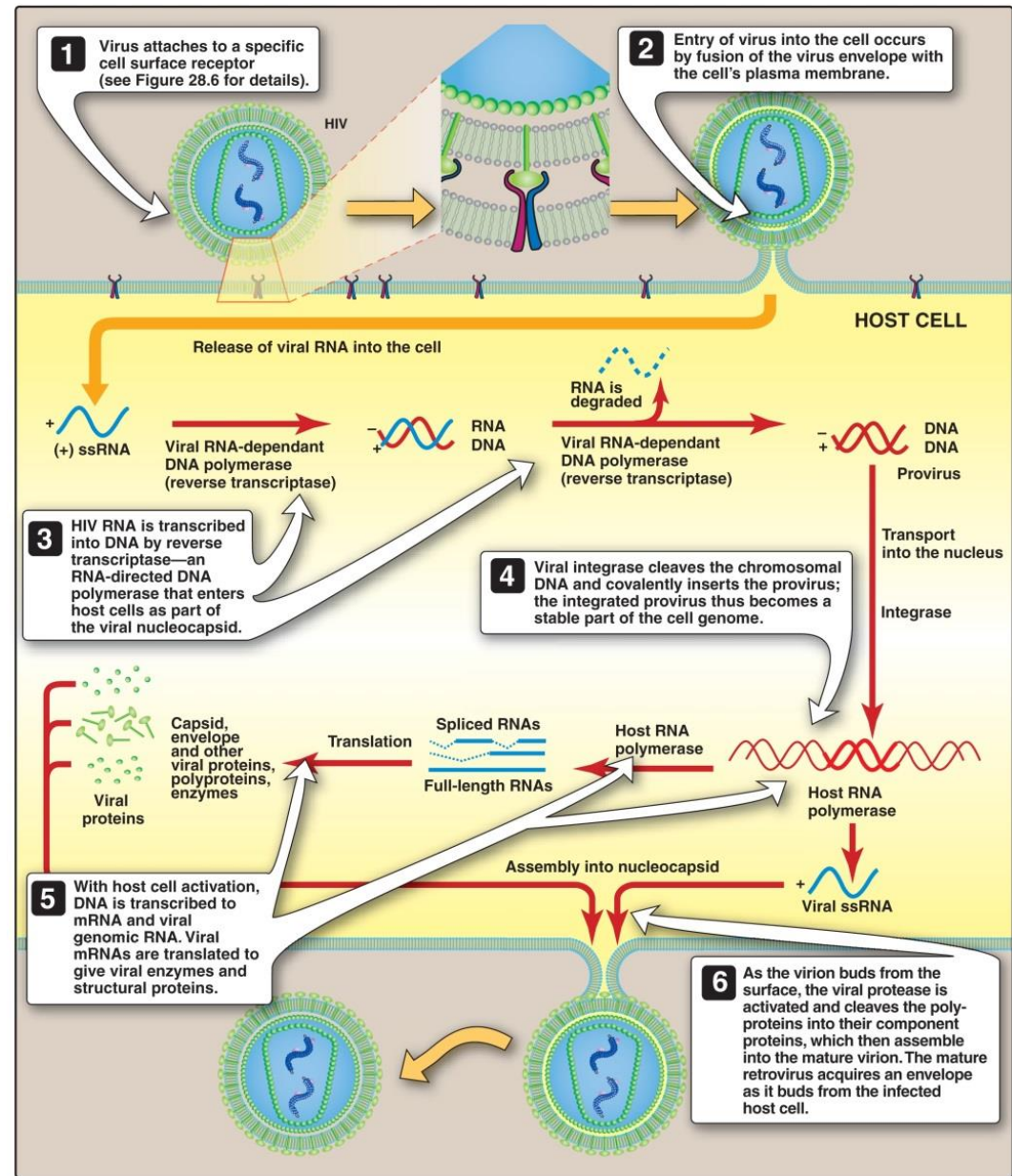
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4788752/>
<https://www.nature.com/articles/nrmicro817>



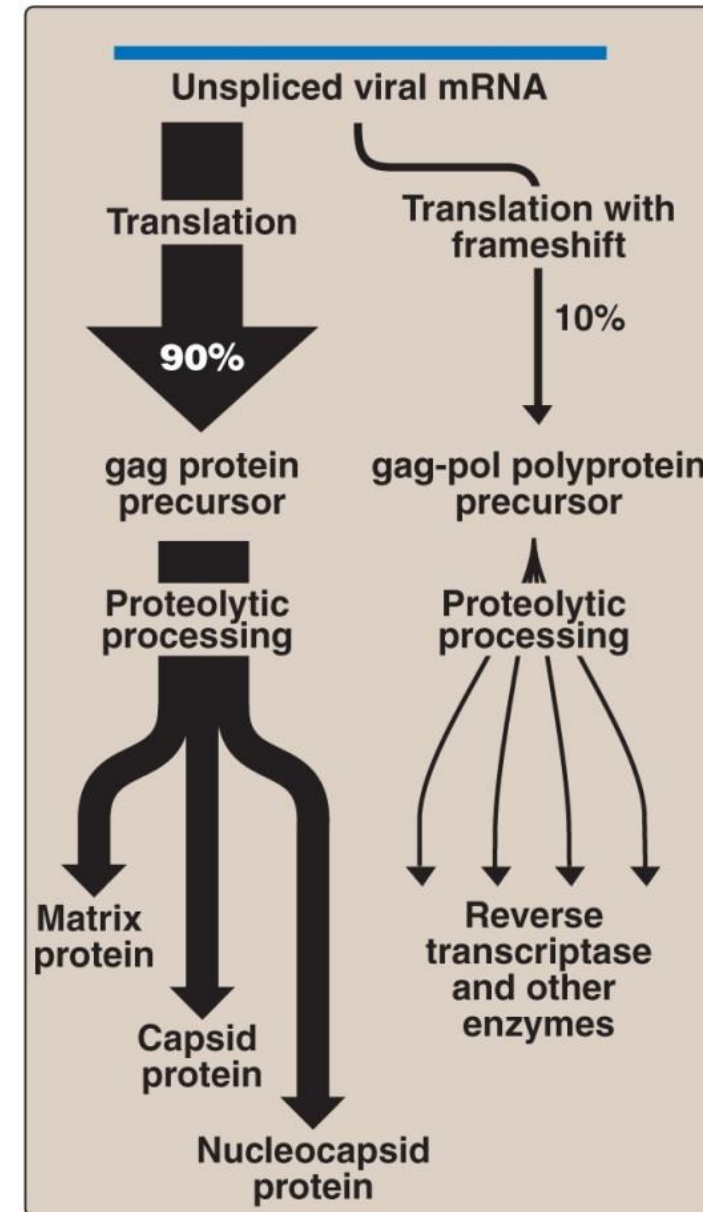
HIV infection



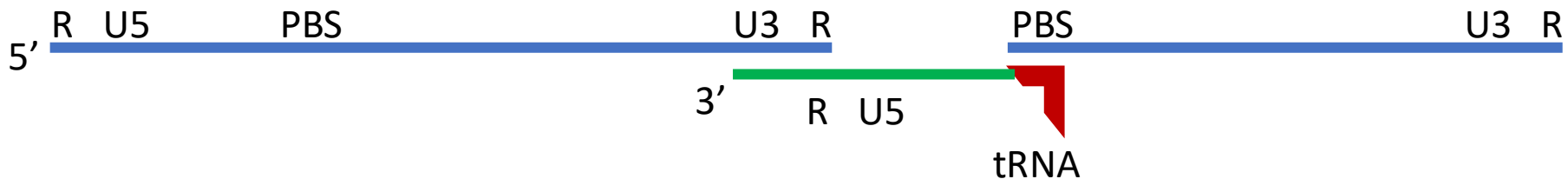
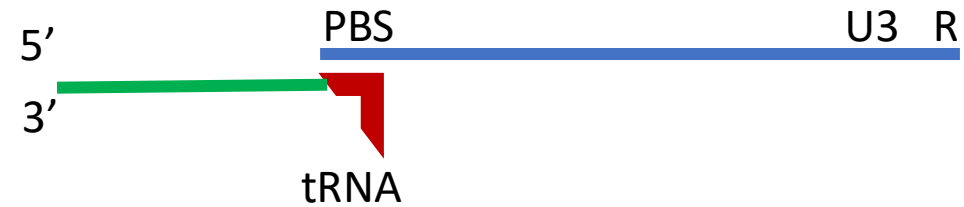
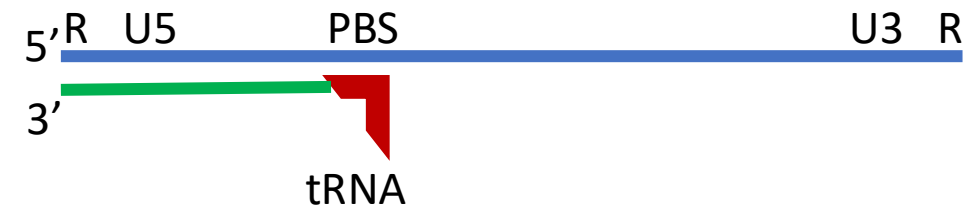
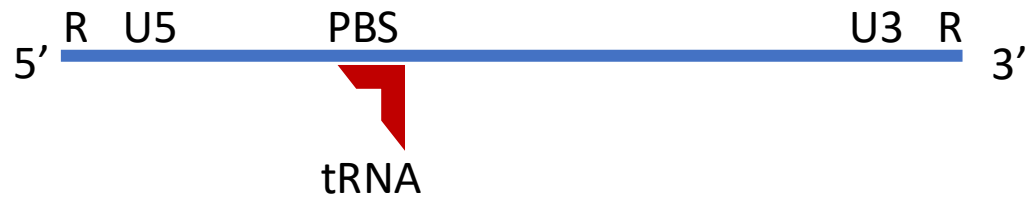
HIV infection



HIV Infection/replication

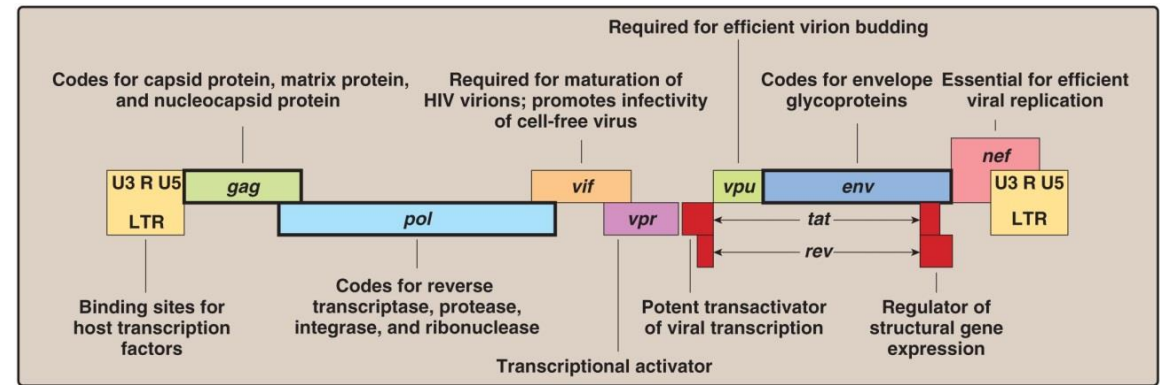


Virus replication

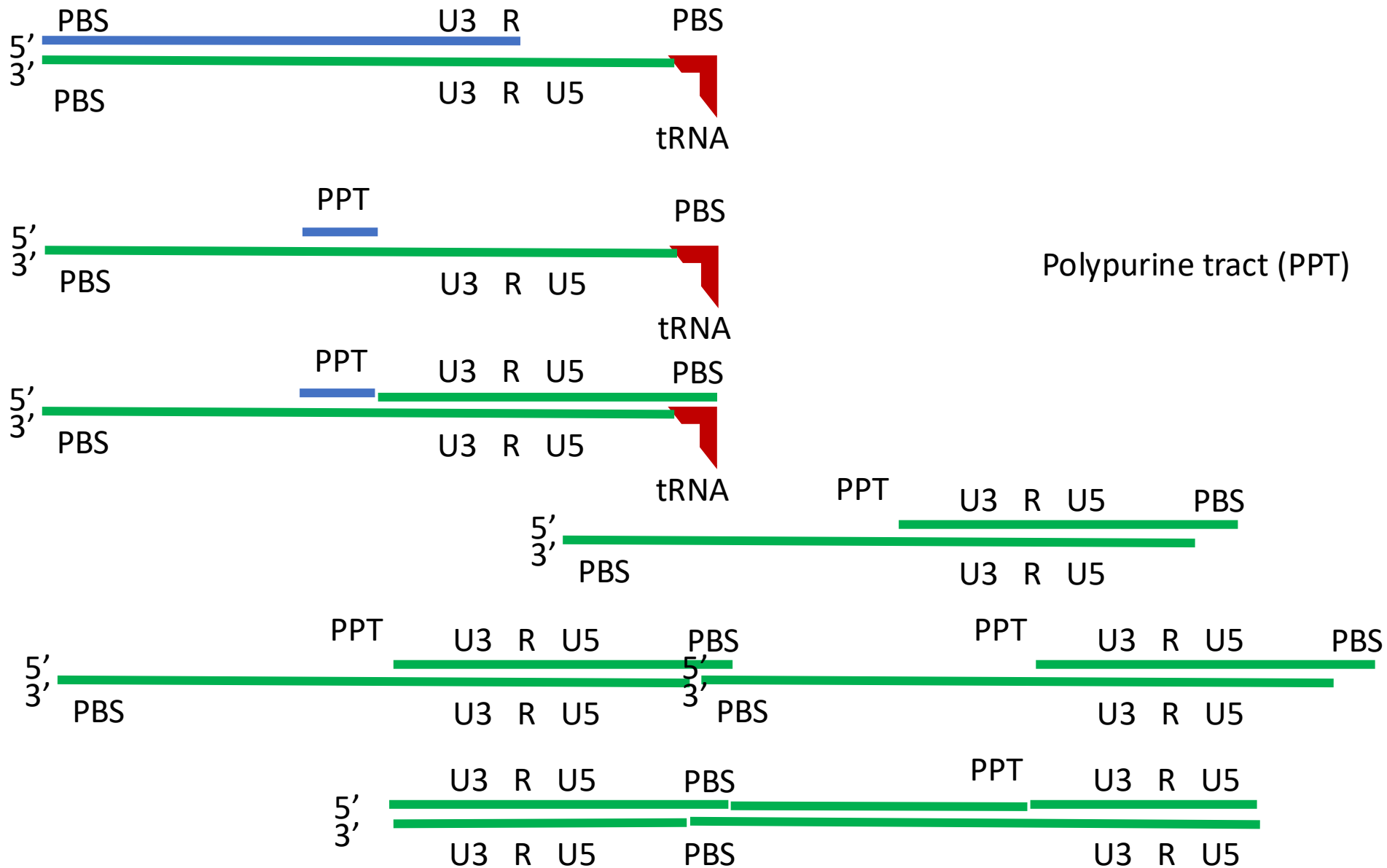


Primer binding site (PBS)

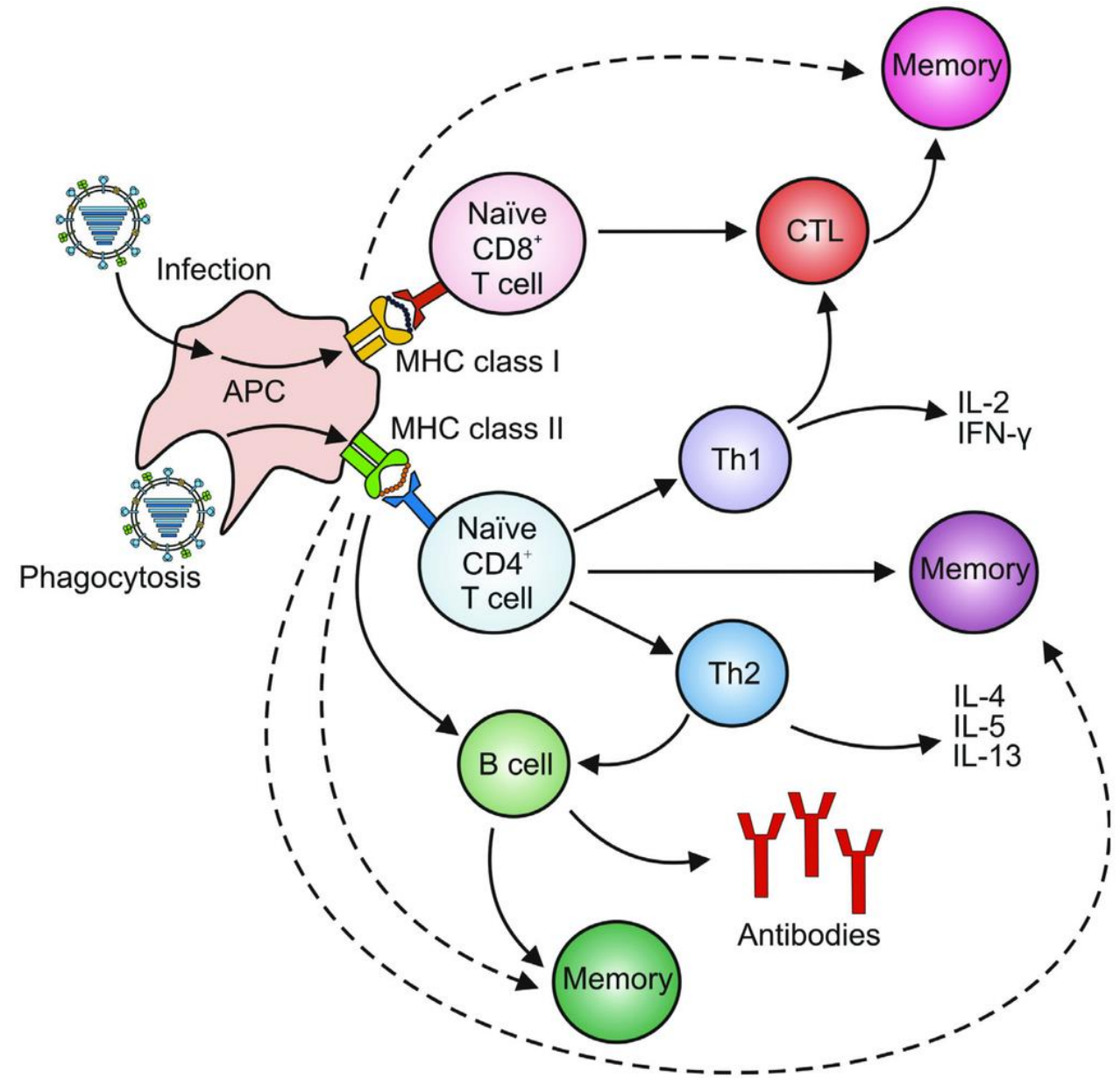
Blue is RNA, green is cDNA



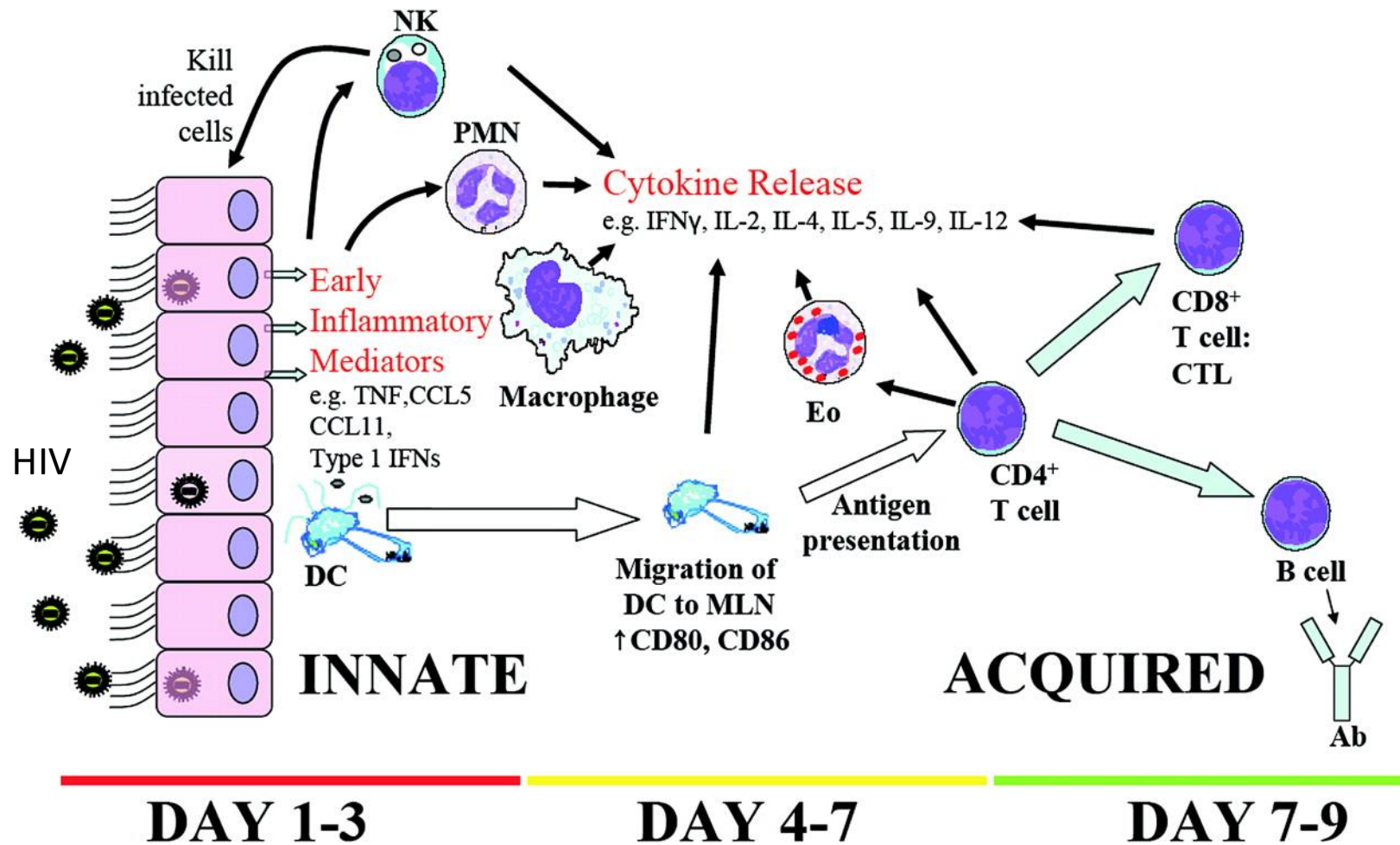
Virus replication



Immune response



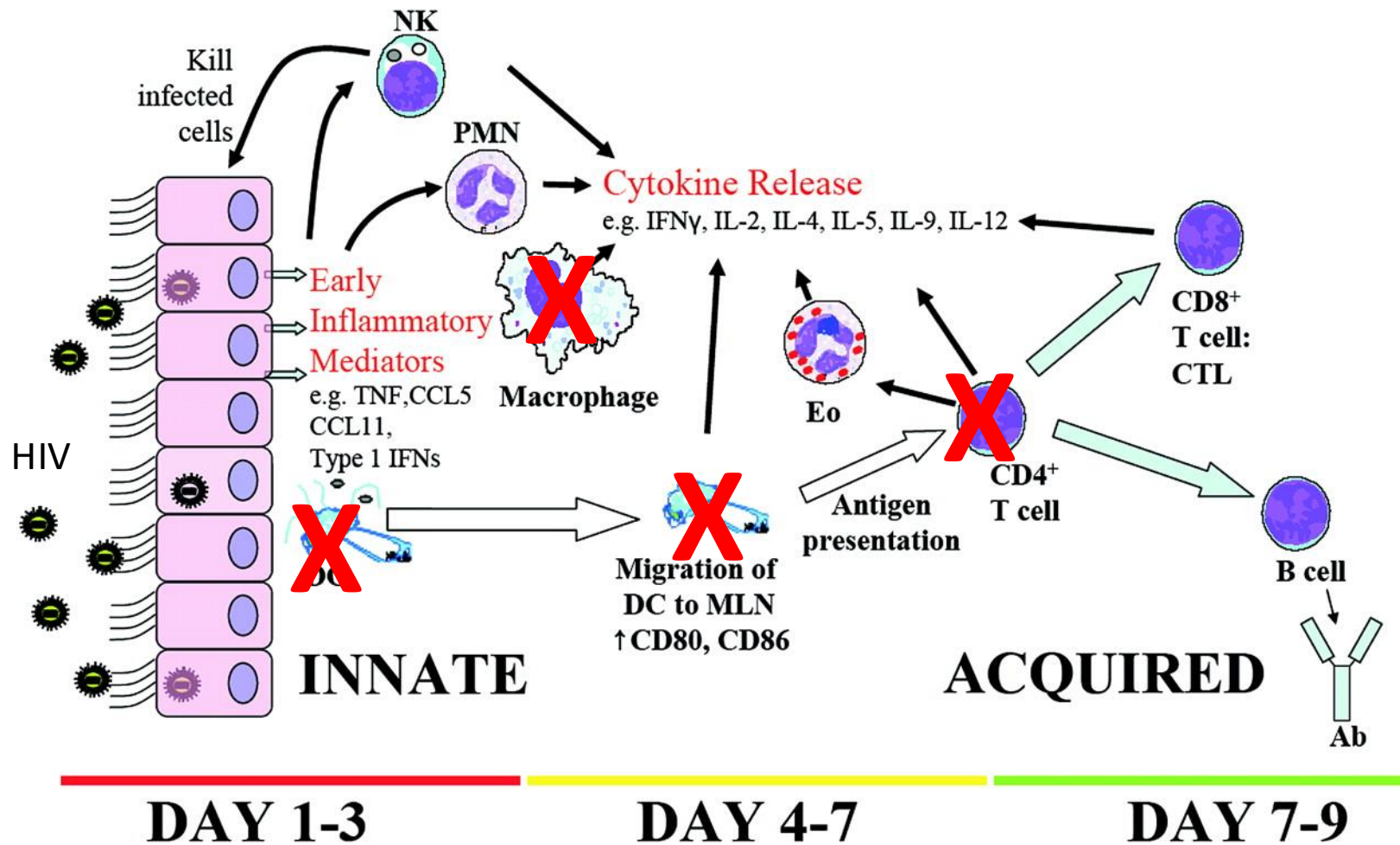
Cells involved in the immune response to HIV. Cellular infection triggers the release of early inflammatory mediators, e.g., TNF and IFN- α/β .



Peter J. M. Openshaw, and John S. Tregoning Clin.
Microbiol. Rev. 2005;18:541-555

Clinical Microbiology Reviews

Cells involved in the immune response to HIV. Cellular infection triggers the release of early inflammatory mediators, e.g., TNF and IFN- α/β .



Peter J. M. Openshaw, and John S. Tregoning Clin. Microbiol. Rev. 2005;18:541-555

Clinical Microbiology Reviews



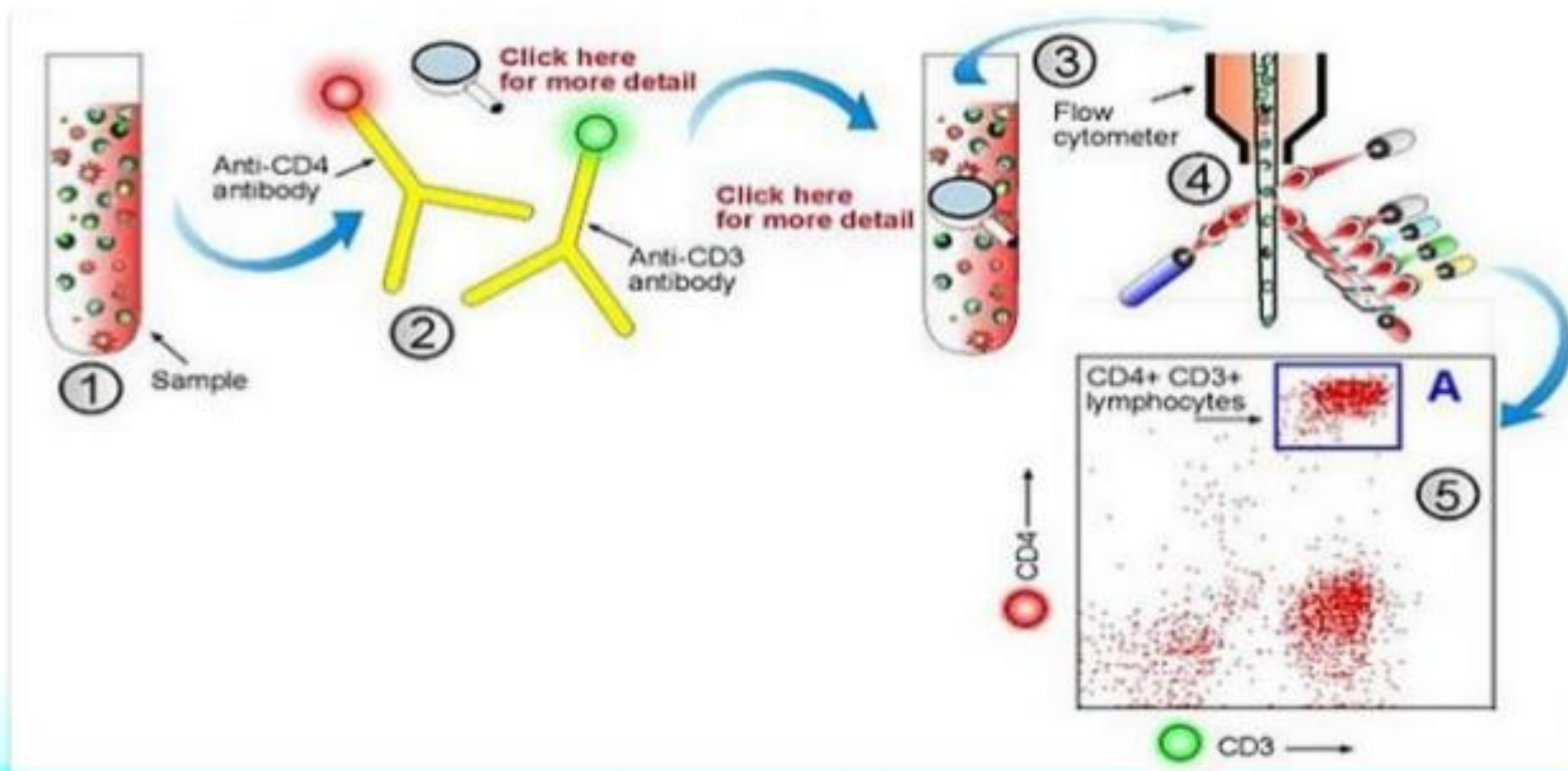
Diagnosis

- Antibody test to HIV antigens in the blood
 - These antibodies can not be detected in the first 4 weeks after infection
- Direct HIV antigen detection in the blood of people
 - This assay is not very sensitive direct after infection, since it is dependent on how much virus is in the body.
 - RT-qPCR for HIV genome.
- AIDS
- Tested by CD4 T cell count by flow cytometry long term



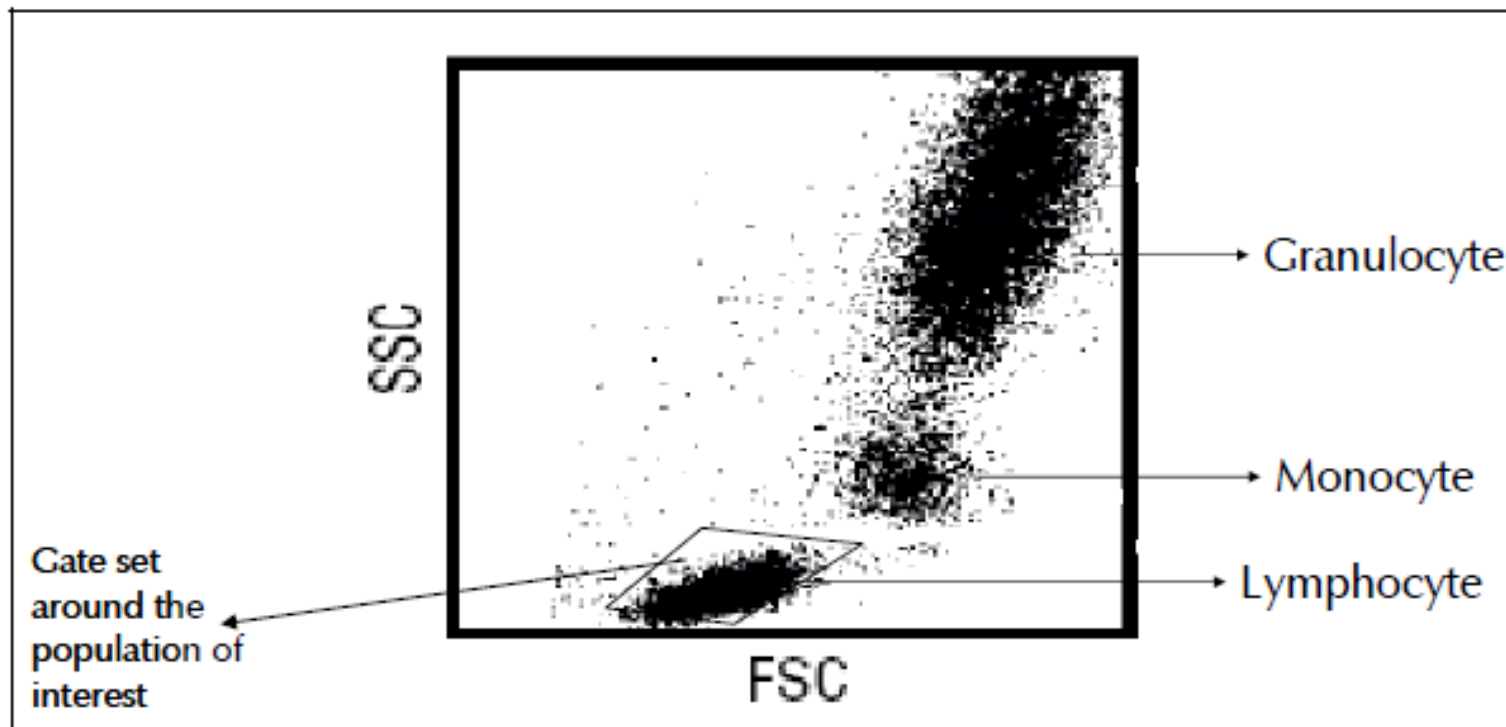
CD4 T cell counts

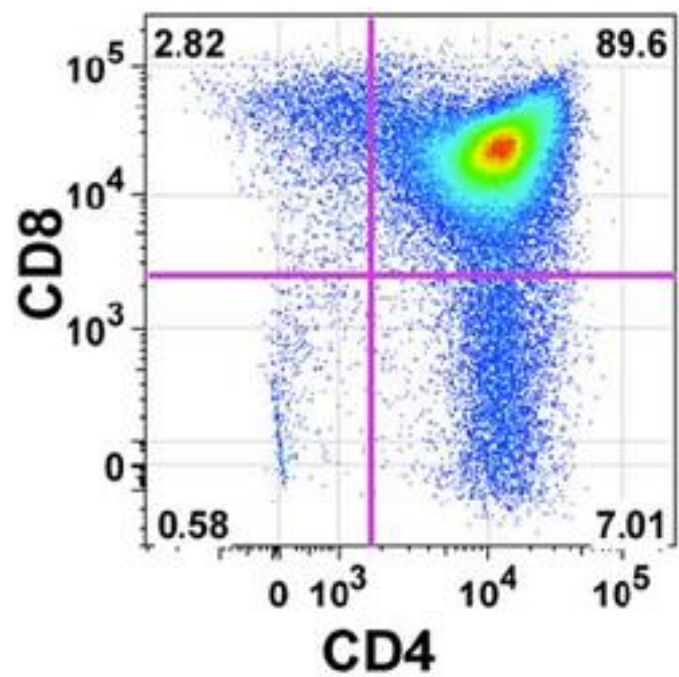
Methods: Flow cytometry



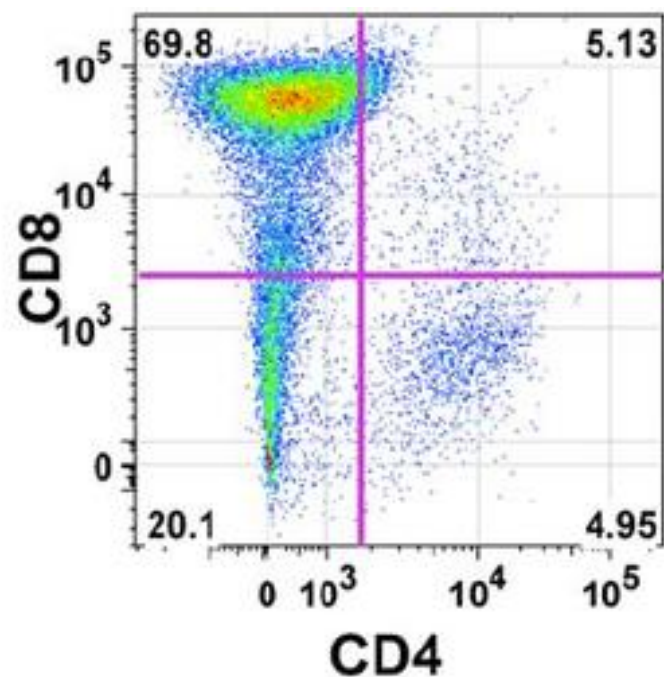
Quantifying the CD4 T cell count

Figure 4.1: Lymphocyte gating using forward scatter and side scatter parameters

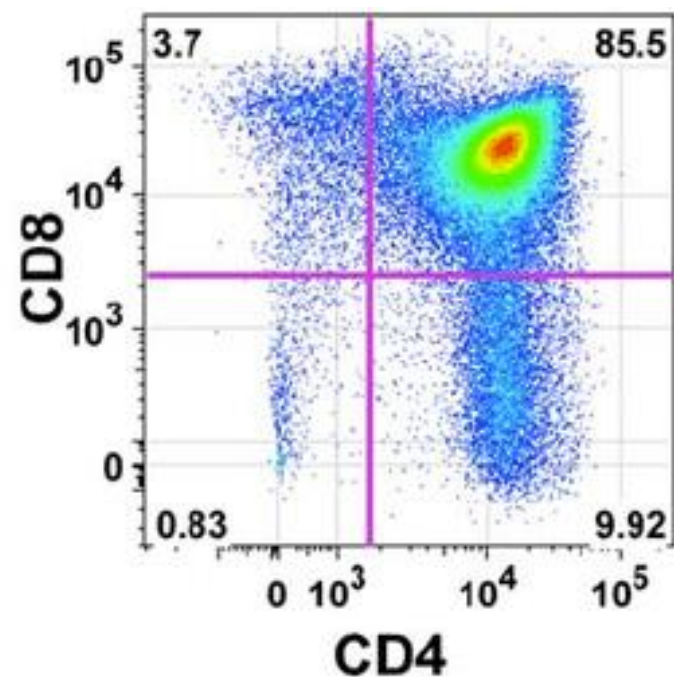




Mock infected

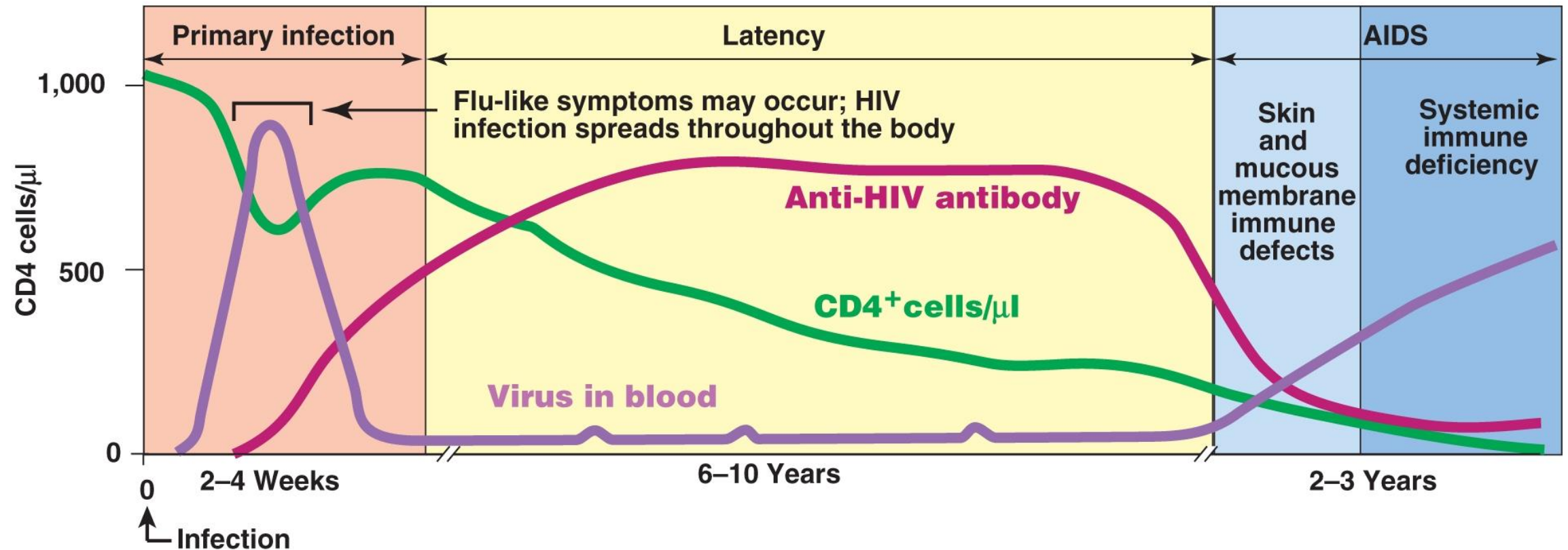


Untreated



3TC (300 mg/kg/day)

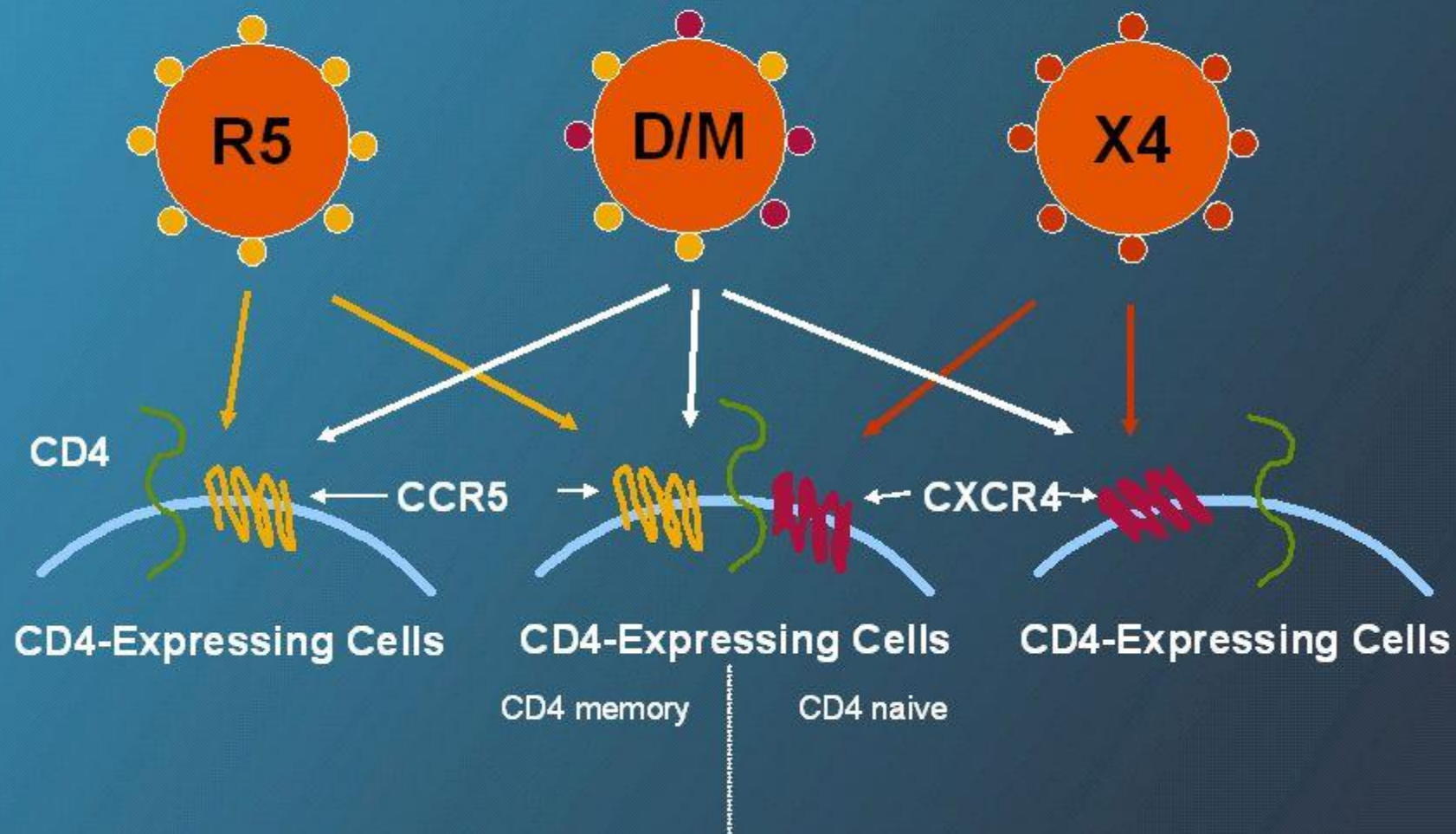
From HIV to AIDS



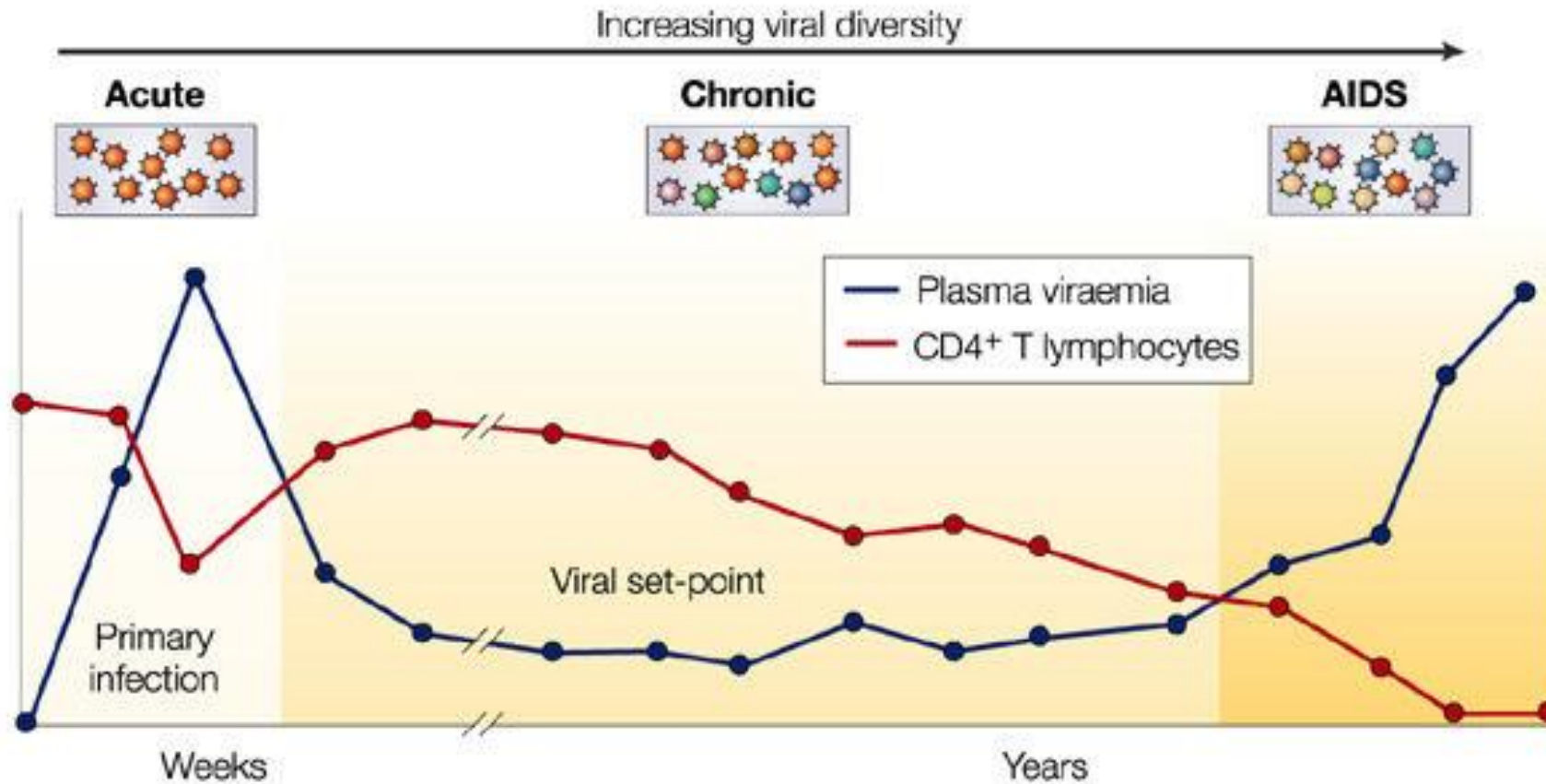
Copyright © 2013 Wolters Kluwer Health | Lippincott Williams & Wilkins

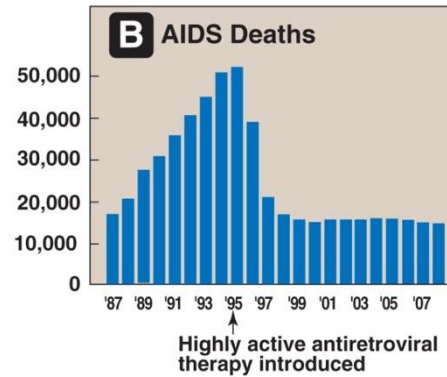
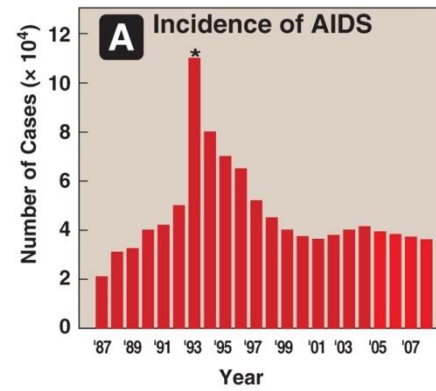
CD4 count ranging from 500 to 1,600 cells per cubic millimeter of blood (cells/ul)
AIDS when the CD4 T cell count is less than 200 cells/ul

Co-receptor Usage of HIV-1 Variants: Defining Tropism

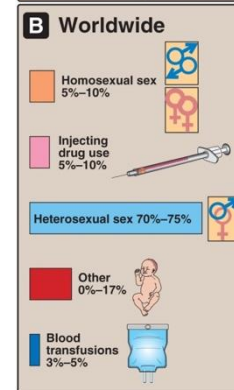
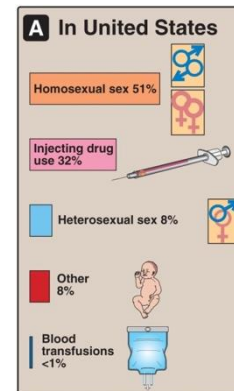


Viral diversity/tropism in time

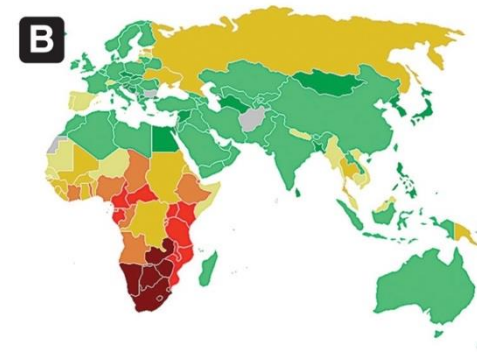
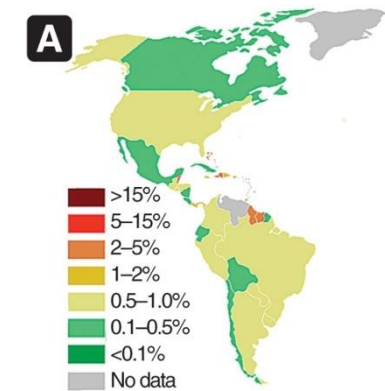




Copyright © 2013 Wolters Kluwer Health | Lippincott Williams & Wilkins



Copyright © 2013 Wolters Kluwer Health | Lippincott Williams & Wilkins

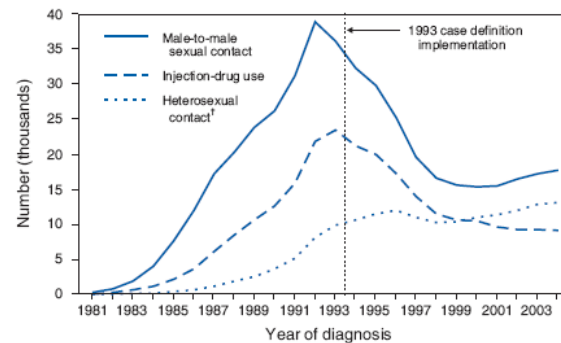


Copyright © 2013 Wolters Kluwer Health | Lippincott Williams & Wilkins

HIV epidemiology

Epidemiology

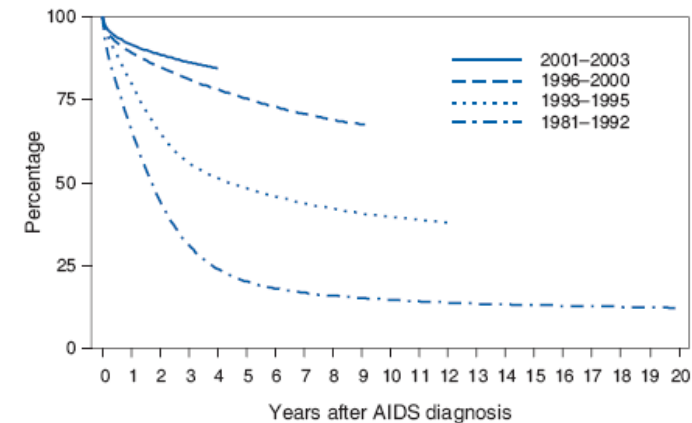
FIGURE 1. Number of acquired immunodeficiency syndrome (AIDS) cases, by major transmission category and year of diagnosis — United States, 1981–2004*



* Data adjusted for reporting delays. Cases without an assigned transmission category were redistributed on the basis of historical trends in risk factors.

† Defined as sexual contact with a person at high risk for or infected with HIV.

FIGURE 2. Percentage of persons surviving through June 2005, by years after acquired immunodeficiency syndrome (AIDS) diagnosis cohorts during 1981–2003 and by year of diagnosis — United States



- 2.1 million (1.8 million–2.4 million) were infected in 2015
- 1.1 million (940 000–1.3 million) people dies of aids in 2015

Globally, **33.4 million** people
are living with HIV/AIDS.

25 million have died since **1981**.

97% of people with
HIV live in developing and
moderate-income nations.

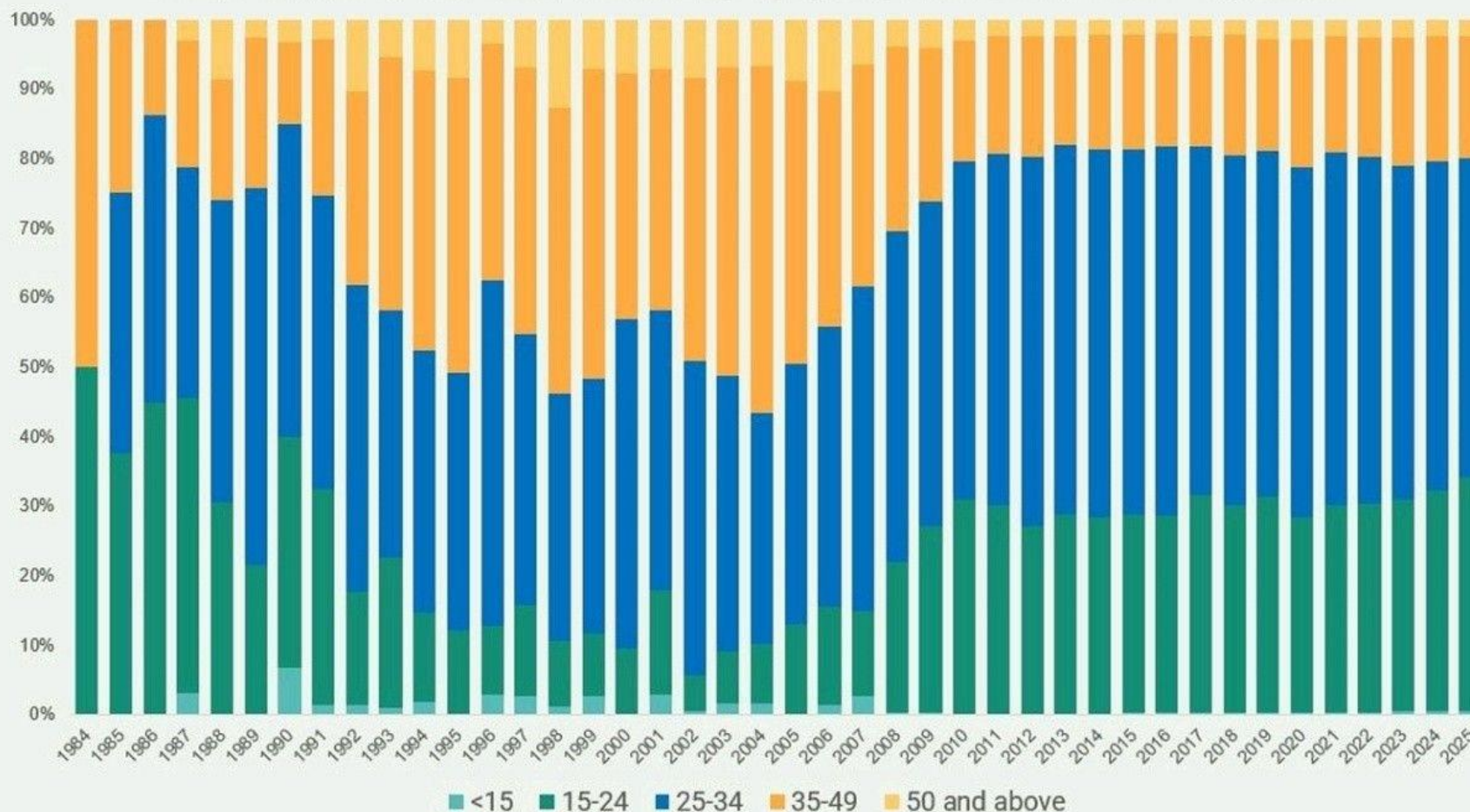


28.6 million people in
poor and moderate-income
countries should be on
antiretroviral medication.

Only **1/3** of these people are
getting this treatment.

Diagnosed HIV cases are getting younger. The predominant age group shifted from among 35-49 years old in 2002 to 2005, to 25-34 years old from 2006 onwards.

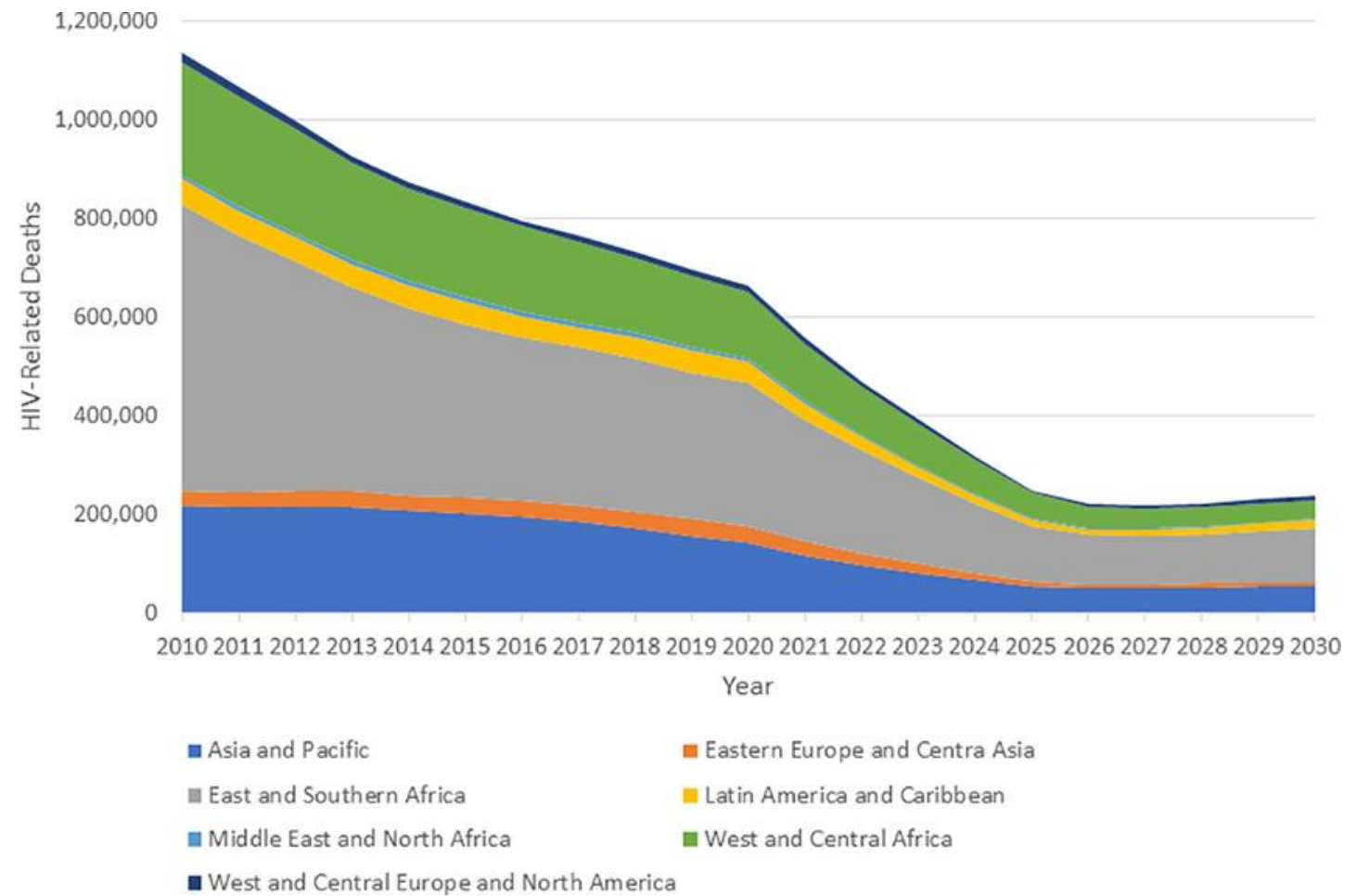
Distribution of diagnosed HIV cases, by age group at diagnosis, 1984 - Apr 2025



Age at Diagnosis

- 2002 - 2005:
44% were 35-49 years old
- 2006 - present:
50% were 25-34 years old
- **Cumulatively since 1984,**
80% of diagnosed cases
were among those below
35 years old.
 - 50% among 25-34 years old
 - 30% among 15-24 years old
 - <1% among <15 years old

HIV Deaths





37% of people ages 18 to 64 in the U.S. have reported **receiving an HIV test.**



HIV education is mandatory in 33 states and Washington, D.C.



Transmission rates have declined in the U.S since the **mid-1980's** from **130,000** per year to **50,000**.



Modern-day **antiretroviral therapy** reduces the chances of an HIV-positive person transmitting the virus by **96%**.



Condom use remains an inexpensive, cost-effective first line of defense against HIV.

90%
Reduced Risk

A new pill known as **Truvada**, or **pre-exposure prophylaxis (PrEP)**, can reduce the risk of transmission by more than **90%**.



Transmission rates **increased 12%** from 2008 to 2010 among gay and bisexual men in the U.S.

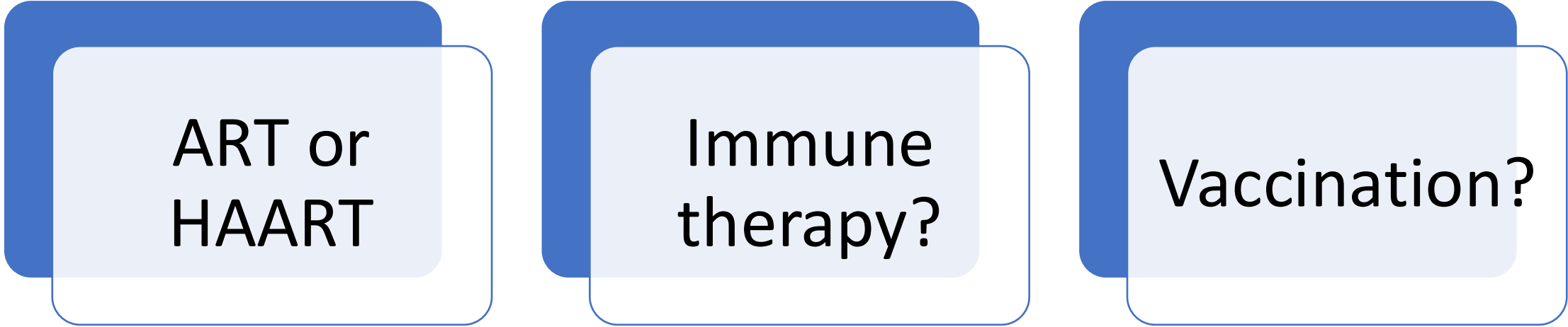
78%

While gay and bisexual men represent only **4%** of the U.S. male population, they comprise **78%** of new HIV infections.

Therapies

- What can we do to treat virus infection?
- Can we get rid of the virus?

Therapies

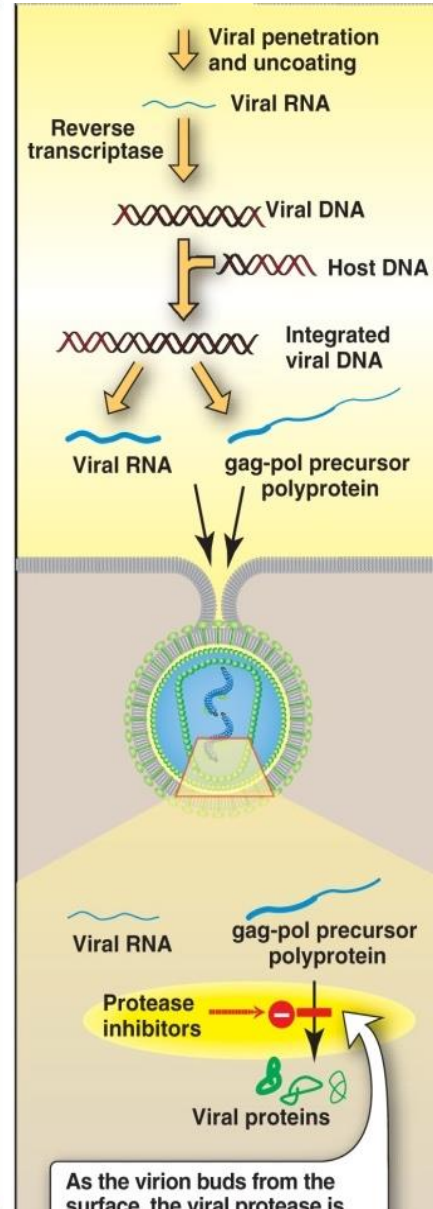
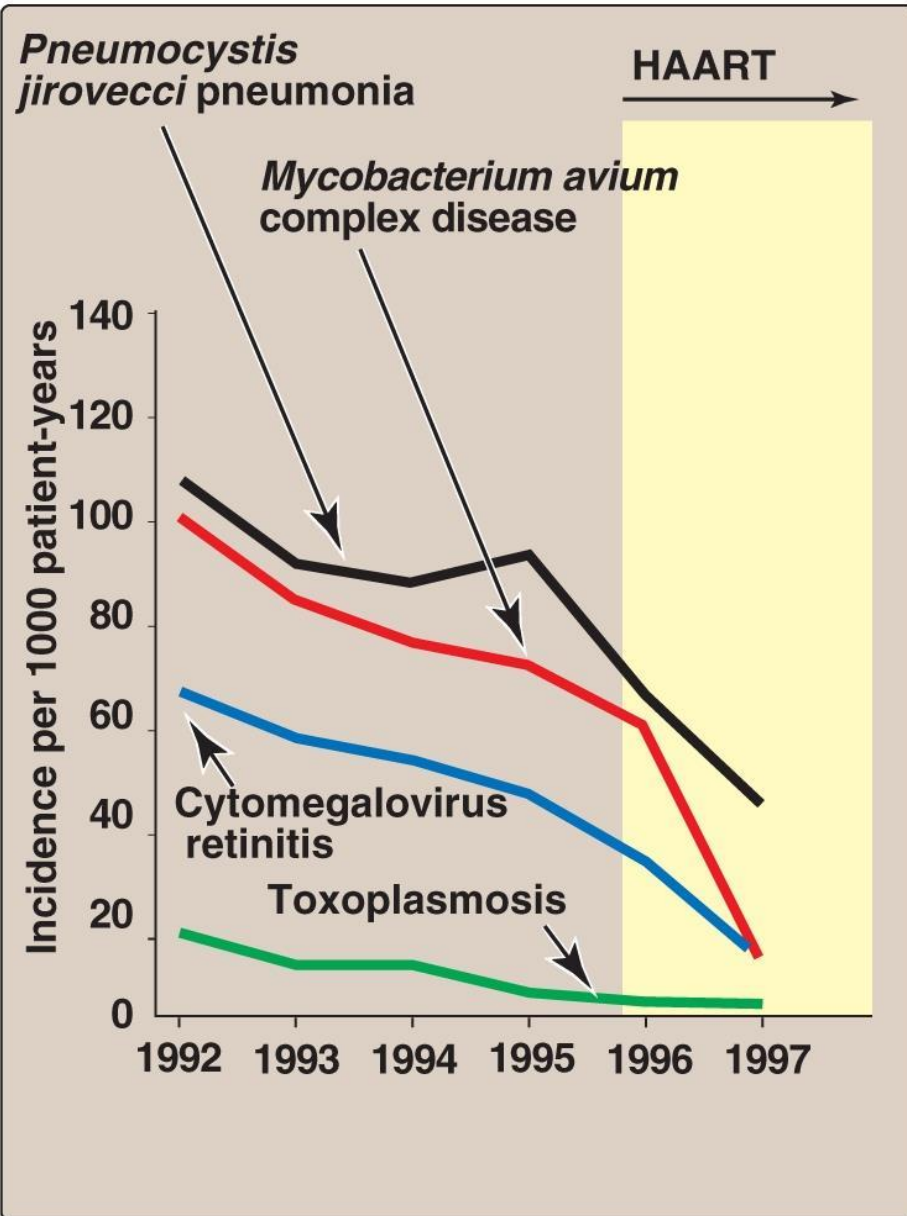


ART or
HAART

Immune
therapy?

Vaccination?

Art or Haart



A Currently available drugs

Nucleoside/-tide reverse transcriptase inhibitors:

- Abacavir
- Didanosine
- Emtricitabine
- Lamivudine
- Stavudine
- Tenofovir
- Zalcitabine
- Zidovudine

Nonnucleoside reverse transcriptase inhibitors:

- Delavirdine
- Efavirenz
- Etravirine
- Nevirapine

Protease inhibitors:

- Amprenavir
- Atazanavir
- Darunavir
- Fosamprenavir
- Indinavir
- Lopinavir
- Nelfinavir
- Ritonavir
- Saquinavir
- Tipranavir

Entry inhibitors:

- Enfuvirtide

Integrase inhibitor:

- Raltegravir

CCR5 antagonist:

- Maraviroc

B Combination therapy¹

Two nucleoside/-tide reverse transcriptase inhibitors

plus

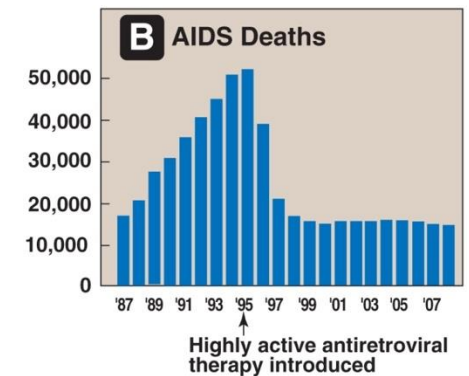
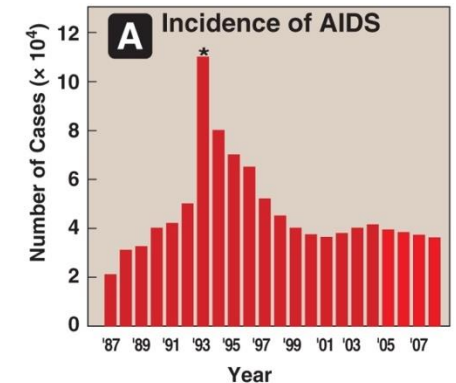
One protease inhibitor (+ ritonavir)

or

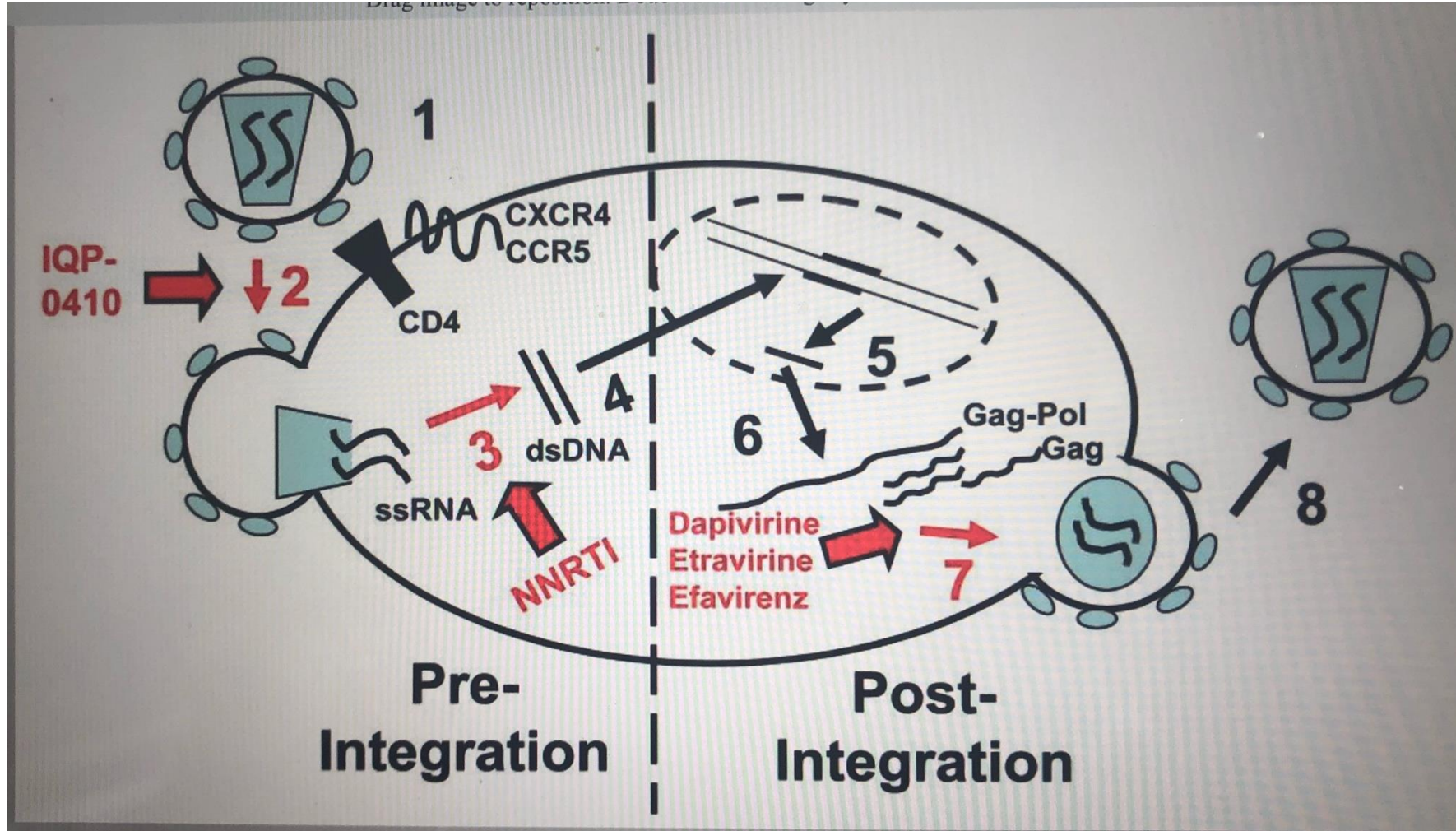
A nonnucleoside reverse transcriptase inhibitor

or

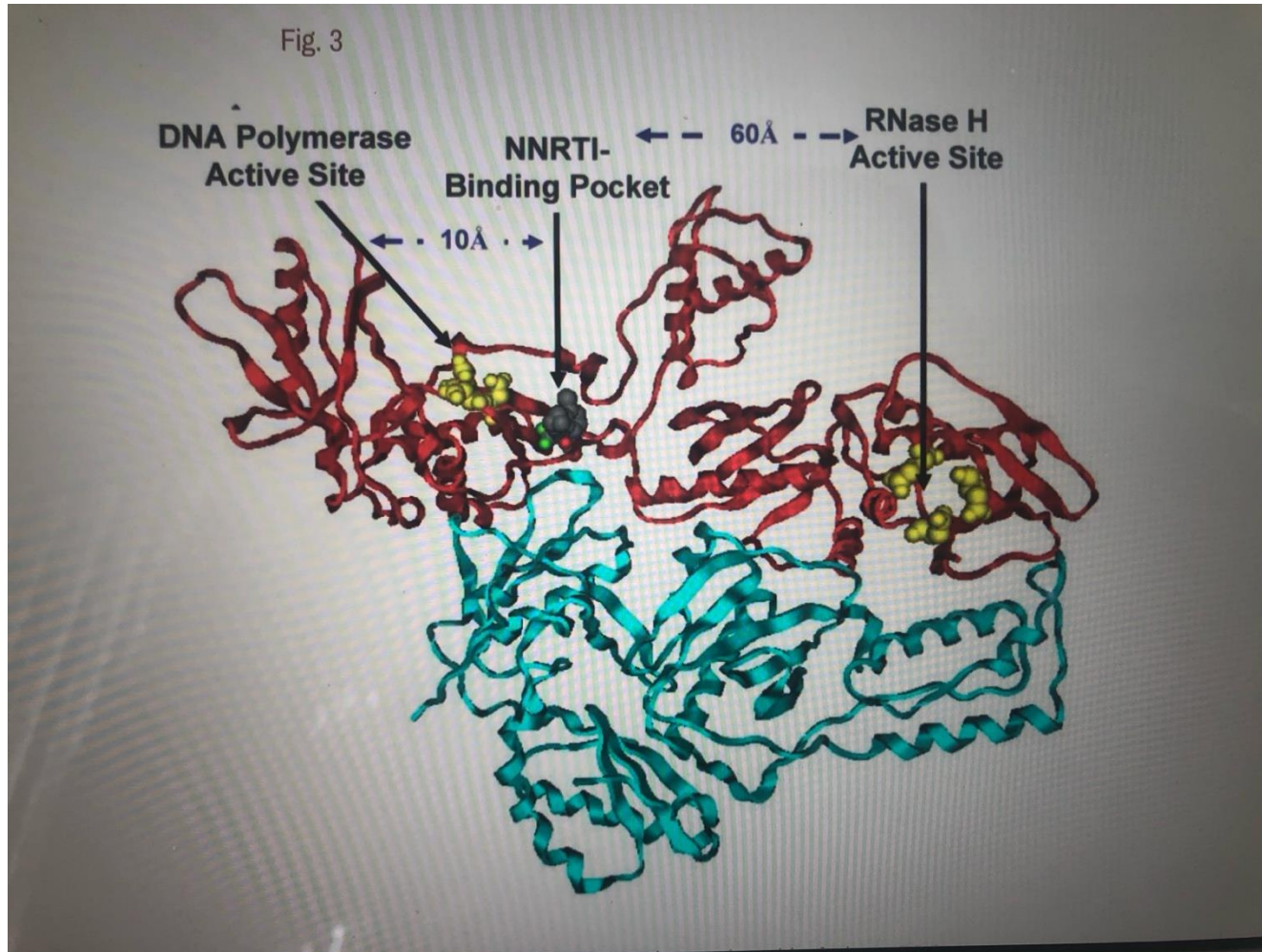
Integrase inhibitor

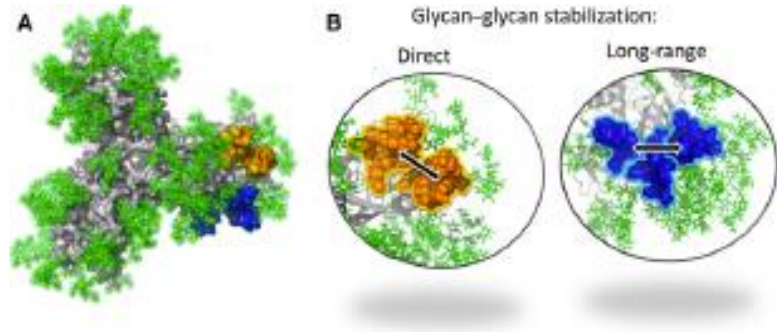
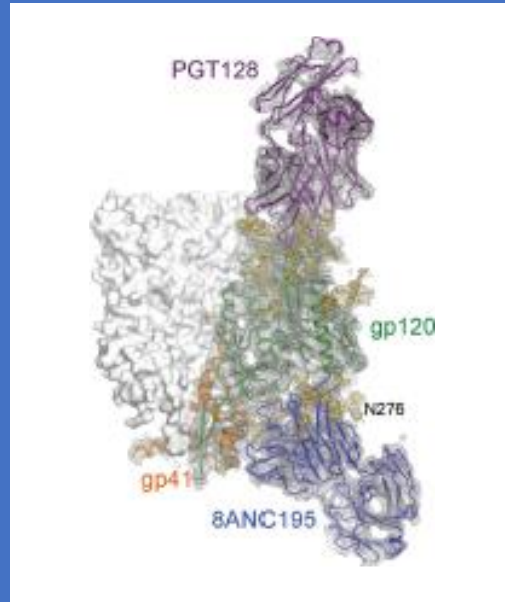


Virus life cycle and inhibitors



Virus life cycle and NNRTI

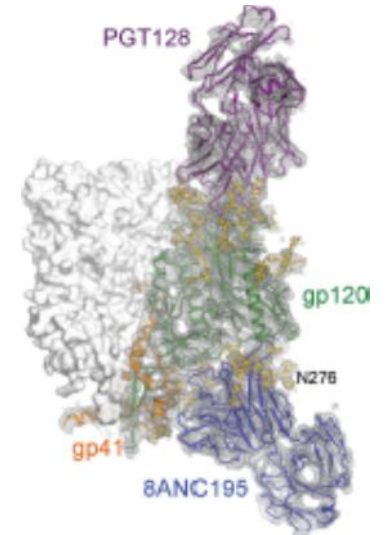




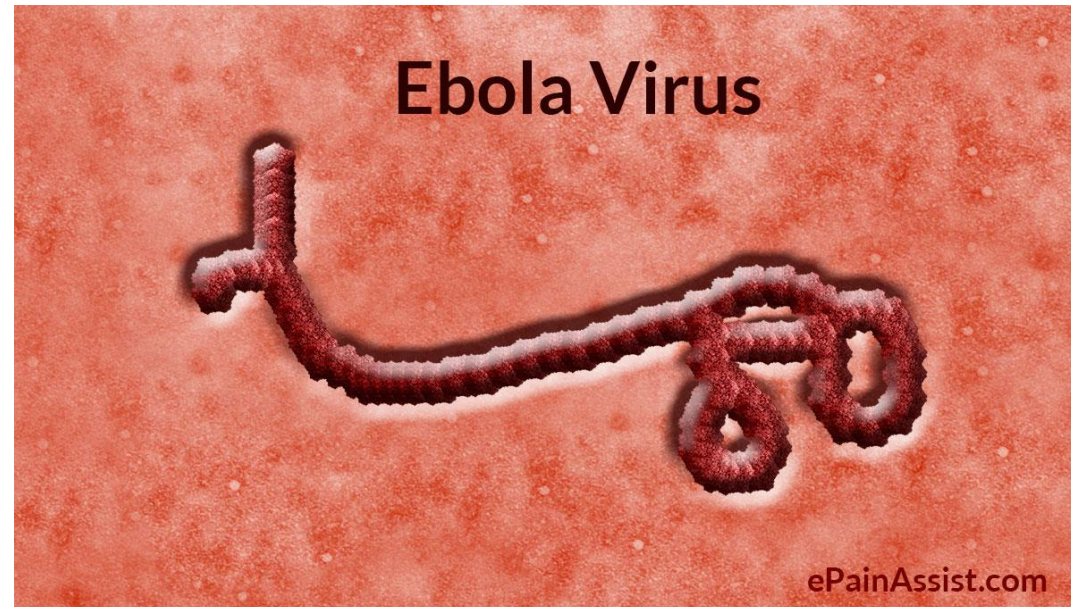
New therapies

Vaccines and antibodies

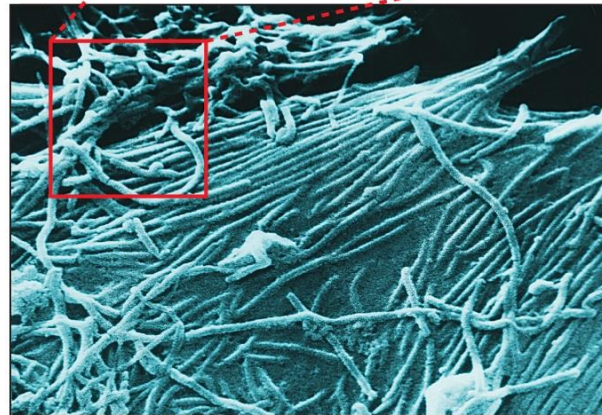
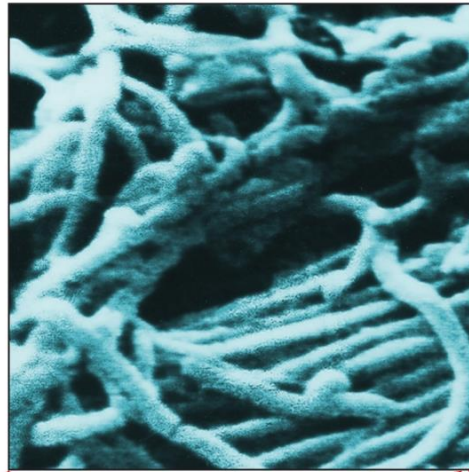
- Several antibodies are able to neutralize HIV virus.
 - Adoptive transfer of these antibodies to infected people or people at risk for infections
- Develop vaccines based on sequences recognized by these antibodies



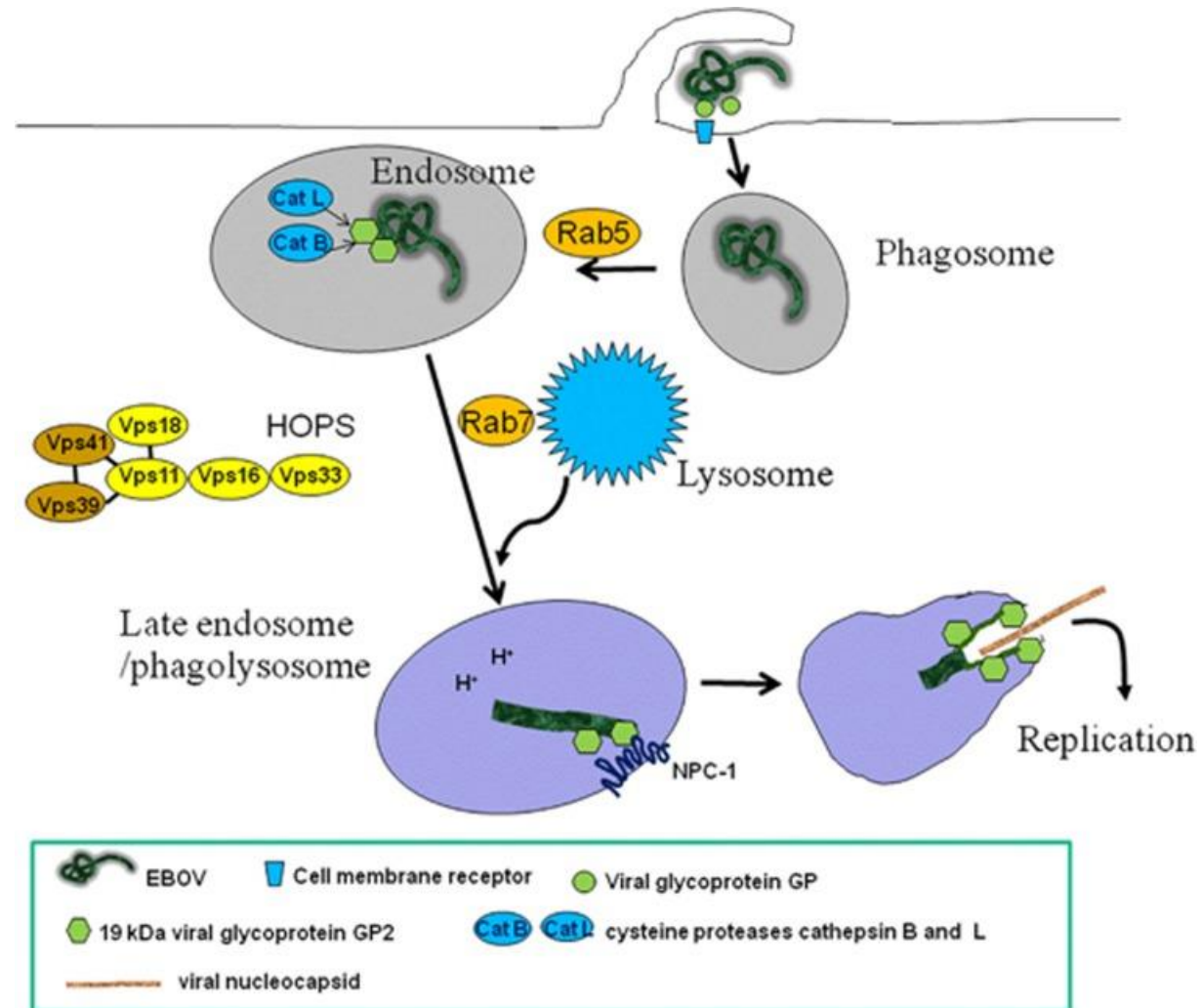
Ebola



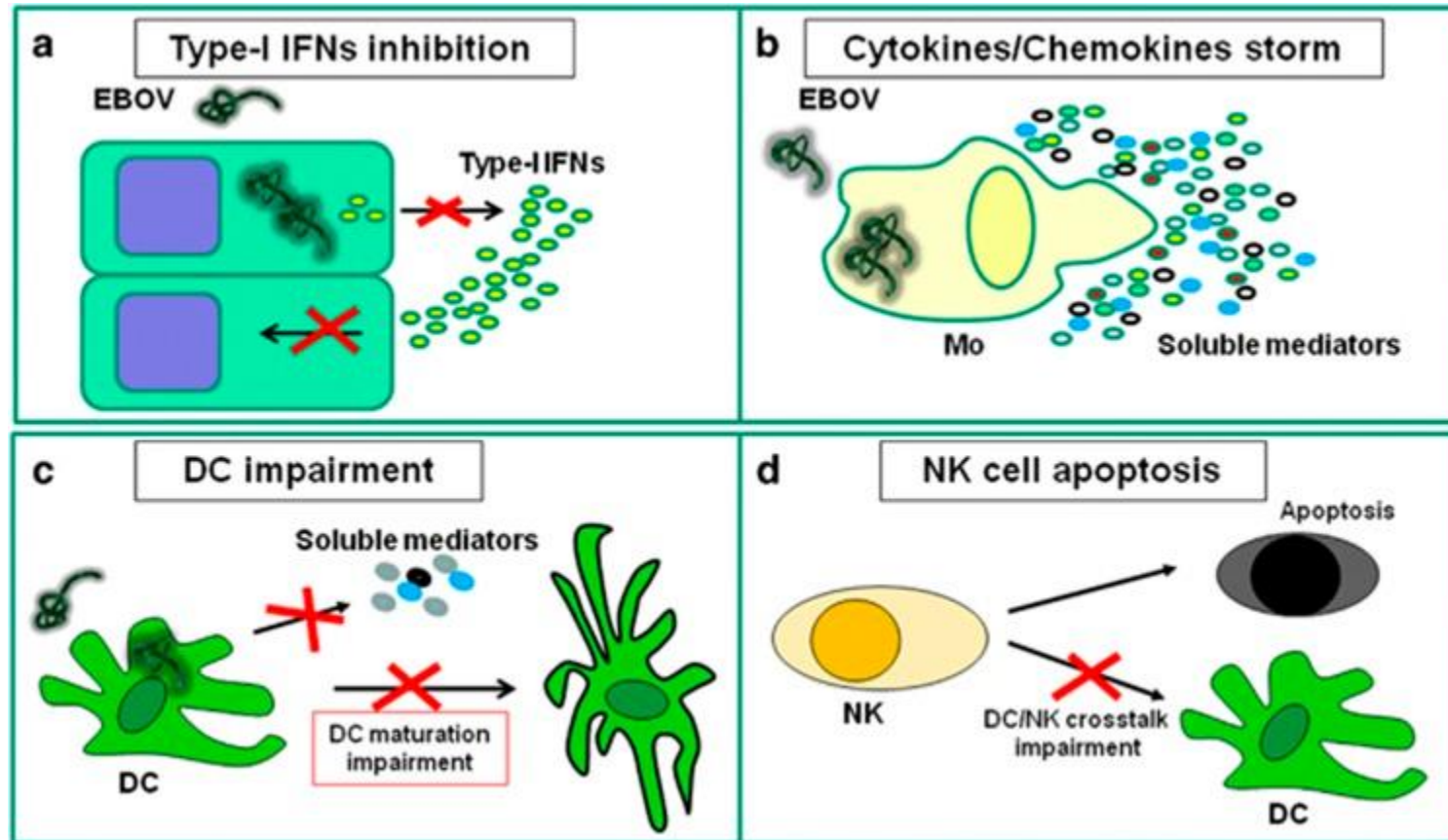
Filoviridae



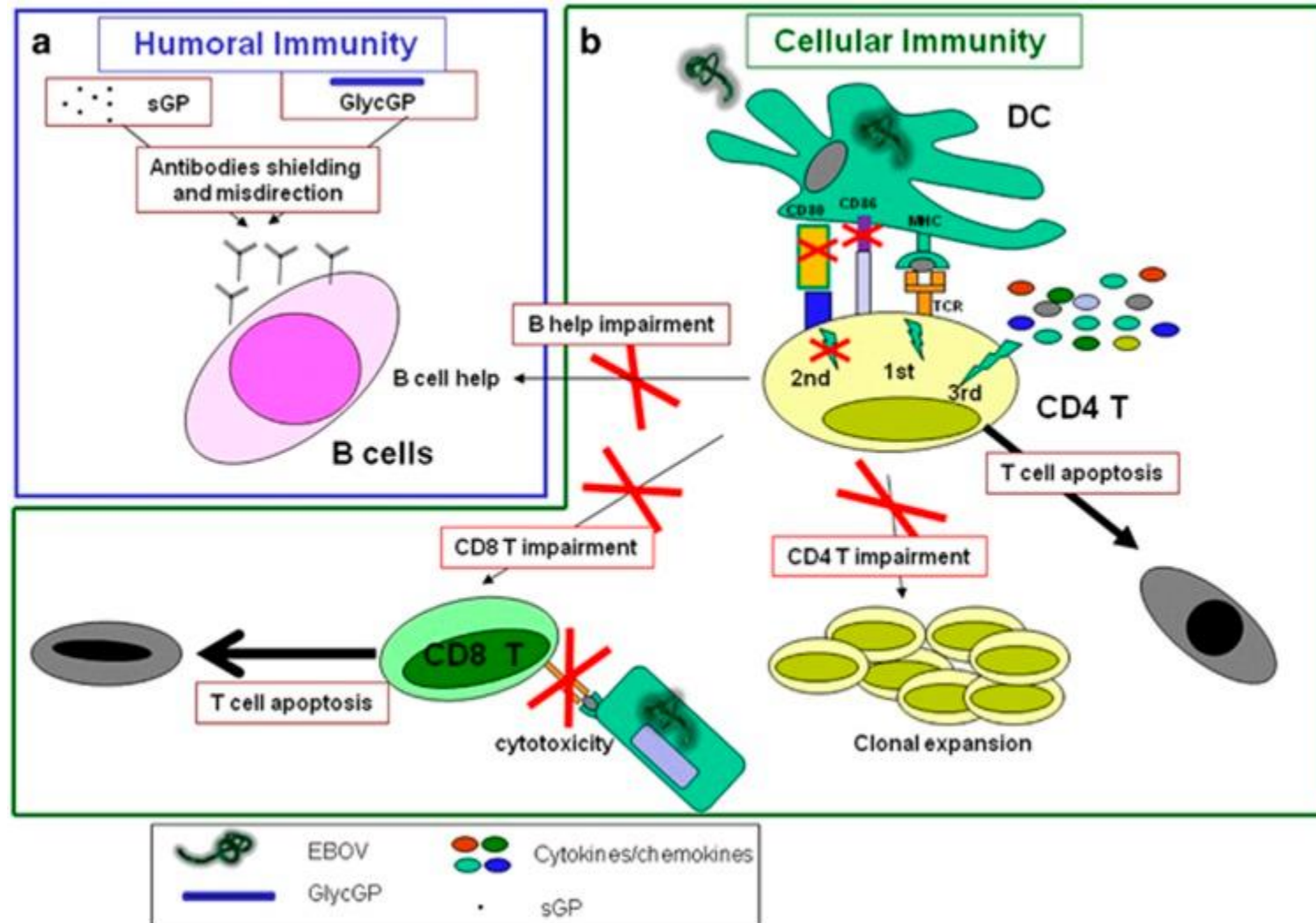
Ebola Infection



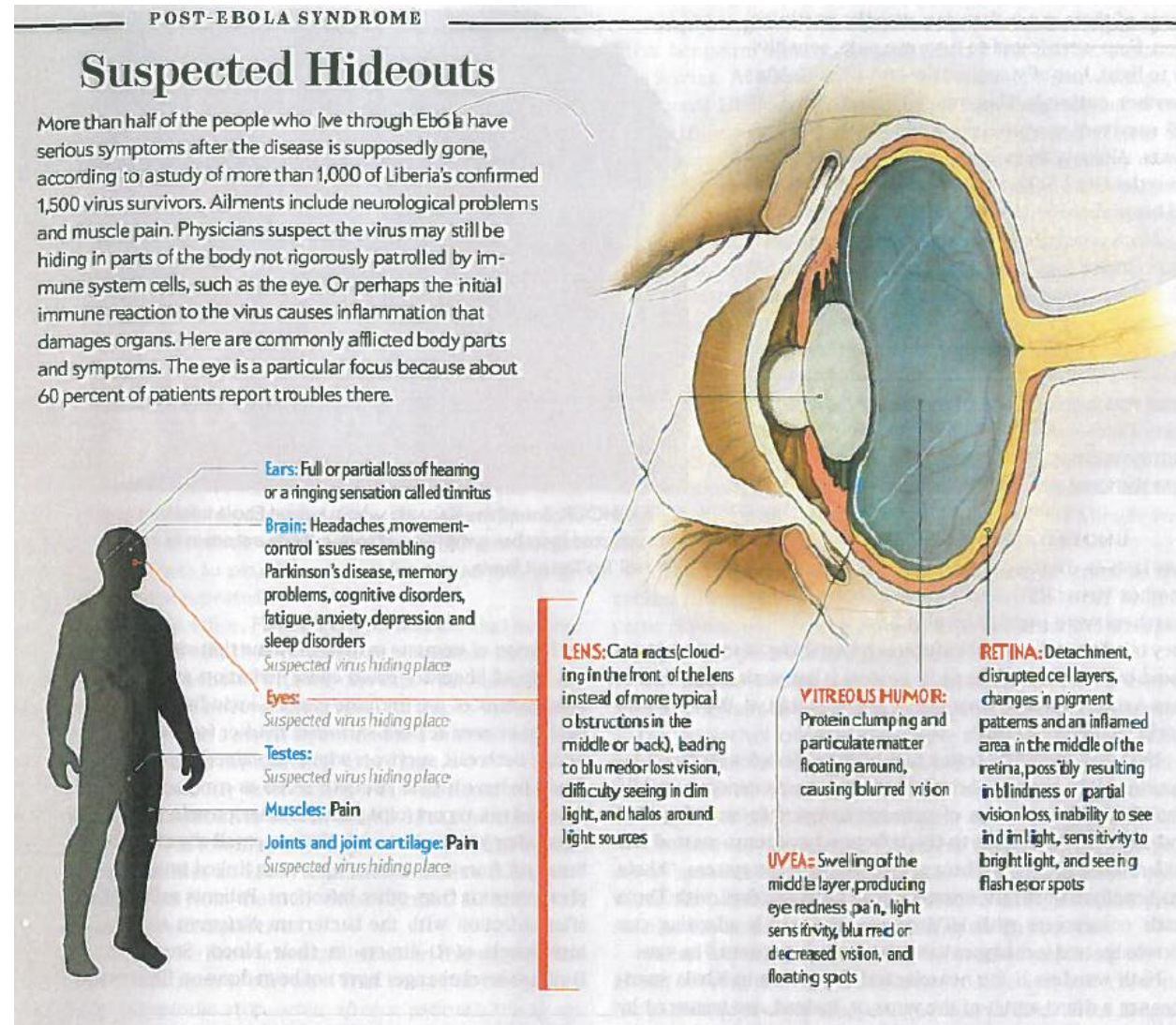
How Ebola blocks the Immune system



How Ebola blocks the Immune system



Ebola Virus hide out





"Mr. Osborne, may I be excused? My brain is full."